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# On a Mathematical Model for One-Dimensional Frustrated Spin Chains: Gamma-Convergence and Numerics

Bachelorarbeit

zur Erlangung des akademischen Grades
Bachelor of Science (B. Sc.)

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eingereicht am: 12.07.2025

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# 1 Introduction

In recent years, the study of low-dimensional magnets has attracted considerable attention within the scientific community. A particular area of this research involves frustrated spin systems, which are of fundamental interest in statistical physics and materials science. Many properties of frustrated systems are still not well understood (see [18, Quantum Magnetism] and [8, Frustrated Spin Systems]). Frustration is introduced to describe a situation where a magnetic spin cannot find an orientation that simultaneously satisfies all the energetic interactions with its neighbors. A common example of frustrated systems with one-dimensional character are edge-sharing chains of cuprates (see [14, Frustrated Spin Systems]). In this thesis we study an energy model of one-dimensional frustrated spin chains. The frustration arises from competing interactions between ferromagnetic (F) nearest-neighbors (NN) and antiferromagnetic (AF) next-nearest neighbors (NNN). This system is called F-AF spin chain model. In such a system the ferromagnetic part encourages adjacent spins to align, while the antiferromagnetic part frustrates this alignment by favoring antiparallel orientations for next-to-nearest neighboring spins (see Fig. 1). We focus on the analysis of this classical spin chain model at its lowest energy state, that is, at zero temperature (see [9] for low-temperature properties).

This thesis is mainly based on the paper "Frustrated Ferromagnetic Spin Chains: A Variational Approach to Chirality Transitions" by Cicalese and Solombrino [5]. The structure of this thesis is as follows. We begin reviewing fundamental mathematical concepts in Section 2, intended for readers with an undergraduate-level background in mathematics. The theory of $\Gamma$-convergence is introduced in Section 3 from the ground up. In Section 4 the main analysis of the F-AF model is presented and all model-specific notations are introduced. In Section 5 we discuss an extended spin chain model. Finally, the thesis concludes with a summary of the main results.

The primary contributions of this thesis are threefold. First, we elaborate on theorems and proofs in [5], providing detailed arguments for the underlying $\Gamma-$convergence framework. Second, we perform an extensive numerical validation of the results with illustrations provided throughout the text. Third, as an extension to the original model, we consider a higher-order spin chain model by extending the energy model with a next-to-next-to-nearest neighbor (NNNN) interaction.

It follows a brief overview of the mathematical energy model. Let us consider a spin system over the one-dimensional interval $[0, 1]$. The state $u$ of the system is represented by a spin variable defined on the points of the lattice $\mathbb{Z}_n = \{i \in \mathbb{Z} \mid \lambda_n i \in [0, 1]\}$, where $\lambda_n$ is a small parameter with $\lambda_n \to 0$ as $n \to \infty$. The energy for a given state $u : i \in \mathbb{Z}_n \mapsto u^i \in S^1$, with $S^1$ denoting the unit sphere in $\mathbb{R}^2$, is

$$E_n(u) = -J_1 \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + J_2 \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}),$$

where $J_1 > 0$ is the NN, $J_2 > 0$ the NNN interaction parameter and $(\cdot, \cdot)$ denotes the

inner product between vectors in $\mathbb{R}^2$. We impose periodic boundary conditions, identifying $(u^0, u^1)$ with $(u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor})$, to avoid boundary artifacts in the evaluation of the energy sums. The frustration of this system depends on the relative strength of the F/AF interaction parameters. As a consequence, the frustration is measured in terms of the interaction ratio $\alpha = J_1/J_2$. We aim to minimize this energy as those states are naturally more favorable. Throughout this analysis three primary ordered states of the spin chain appear: ferromagnetic, antiferromagnetic and helimagnetic (H) states. The characteristic arrangement of a one-dimensional spin chain each in a particular state is visualized in Fig. 1. Notably, helimagnetic states exhibit a chirality symmetry, admitting both clockwise and counter-clockwise spin configurations along the chain, which correspond to negative and positive chirality respectively. This symmetry and its breaking will play a role in the analysis of chirality transitions within the model.

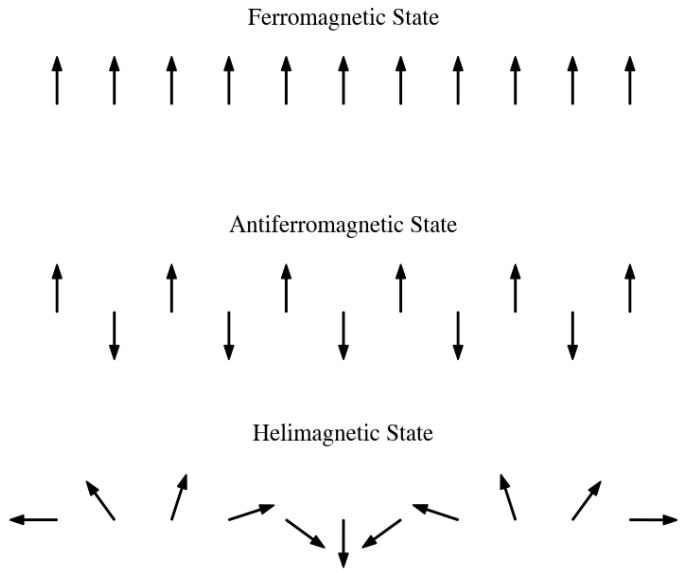


Figure 1: (Top) The ferromagnetic state, where all spins are aligned in the same direction. (Middle) The antiferromagnetic state, characterized by a perfect alternating alignment of adjacent spins. (Bottom) A helimagnetic state, where the spin direction (negative chirality) rotates progressively along the chain to form a spiral structure.

# 2 Notation and Preliminaries

## 2.1 Notation and Function Spaces

**Conventions.** We clarify some general notation. Let $X$ be a normed space. Whenever we write $(x_n)_n \subseteq X$ we refer to a sequence $x : \mathbb{N} \to X$, with $x_n := x(n)$. If the sequence converges for $n \to \infty$ to $\bar{x} \in X$ we write $x_n \to \bar{x}$. The dual space of a normed space, X, is the collection of all bounded linear functionals that map X to its underlying scalar field. The dual space of $X$ is denoted by $X'$.

We denote by $|\cdot| : \mathbb{R}^n \to \mathbb{R}$ the Euclidean norm of an $n$-dimensional vector. All other norms will be specified. We write $\chi_J$ for the indicator function of an interval $J \subseteq \mathbb{R}$.

**$\mathcal{L}^p$ and $BV$.** We write $\Omega \subseteq \mathbb{R}^d$ for a Lebesgue measurable set. By writing $\mathcal{L}^p(\Omega)$ for $p \in [1, \infty]$, we refer to the Lebesgue spaces with the $p$-norm. Sometimes, in addition, we will indicate the range, e.g. $\mathcal{L}^\infty((0,1), \mathbb{R}^2)$. If it is clear from the context, we write the $p$-norm as $\|\cdot\|_p$ and don't mention the domain.

The definition below describes a function space on which some of the functionals live that are computed in Section 4.

**Definition 2.1** (Pointwise Bounded Variation)**.** *[13, Definition 2.1]*
*Let $J \subseteq \mathbb{R}$ be an interval, let (X, d) be a metric space and $f : J \to X$ be a function. A partition of $J$ is a finite set $\{x_0, \ldots, x_N\} \subset J$, where $x_0 < \ldots < x_N$. The pointwise variation of $f$ on $J$ is*

$$Varf := \sup\{\sum_{i=1}^{N} d\left(f(x_i), f(x_{i-1})\right) \mid \{x_0, \ldots, x_N\} \text{ partition of } J\}$$

*A function has finite or bounded pointwise variation if $Varf < \infty$. The space of functions with bounded pointwise variation is denoted by $BV(J, X)$.*

**Remark 2.2.** *If $\sup J$ (or $\inf J$) is finite but not contained in $J$, then by convention the partition points $x_n$ (or $x_0$) are strictly less than $\sup J$ (or strictly greater than $\inf J$). This ensures that the definition of $Varf$ remains well-defined without requiring evaluation of $f$ at endpoints not contained in $J$.*

**Example 2.3.** *Consider the function $f : (0,1) \to \{\pm 1\}$ with standard metric on $\mathbb{R}$ given by $f(x) = \chi_{(0,1/2]} - \chi_{(1/2,1)}$. This function has a single jump discontinuity at $x = 1/2$ and its variation $Varf = 2 < \infty$. Therefore $f \in BV((0,1), \{\pm 1\})$.*

**State Variable Sets** Here, we collect the sets introduced in Sections 3 and 4. They will be used repeatedly. Let the sequence $(\lambda_n)_n$ be such that $\lambda_n > 0$ for all $n \in \mathbb{N}$

and $\lambda_n \to 0$. Set $I := (0,1) \subset \mathbb{R}$.

$$\begin{aligned}
\mathbb{Z}_n(I) &= \{i \in \mathbb{Z} : \lambda_n i \in \bar{I}\}, \qquad (1)\\
\mathcal{U}_n(I, S^1) &= \{u : i \in \mathbb{Z}_n(I) \mapsto u^i \in S^1\},\\
\bar{\mathcal{U}}_n(I, S^1) &= \{u \in \mathcal{U}_n(I, S^1) | u \text{ satisfies } (u^0, u^1) = (u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor})\},\\
\mathcal{C}_n(I, S^1) &= \{u \in \bar{\mathcal{U}}_n(I, S^1) | u(x) = u(\lambda_n i) \text{ if } x \in \lambda_n(i + [0,1)), \text{ for all } i \in \mathbb{Z}_n(I)\},\\
\mathcal{D}_n(I, \mathbb{R}^m) &= \{u : \mathbb{R} \to \mathbb{R}^m \mid u \text{ constant on } \lambda_n(i + [0,1)) \text{ for all } i \in \mathbb{Z}_n(I)\}.
\end{aligned}$$

In the following we recall some results from Functional Analysis. Where no proof is given, it can be found in the given reference.

## 2.2 Convergence and Compactness

### Modes of convergence.

**Definition 2.4** (Weak Convergence)**.** *[16, Definition 27.1] We say that a sequence $(x_n)_n \subseteq X$ converges weakly to $x \in X$ and write $x_n \rightharpoonup x$ if*

$$f(x_n) \to f(x) \text{ for all } f \in X'.$$

**Definition 2.5** (Weak* Convergence)**.** *[16, Definition 27.8] We say that a sequence $(f_n)_n \subseteq X'$ weakly* converges to $f \in X'$ and we write $f_n \overset{*}{\rightharpoonup} f$, if*

$$f_n(x) \to f(x) \text{ for all } x \in X.$$

**Remark 2.6.** *Let $p \in [1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$ with the convention that $\frac{1}{\infty} = 0$. The dual space of $\mathcal{L}^p(\Omega)$ can be identified with $\mathcal{L}^q(\Omega)$, i.e. $(\mathcal{L}^p(\Omega))' \simeq \mathcal{L}^q(\Omega)$. The representation is as follows: for every $g^* \in (\mathcal{L}^p(\Omega))'$ there exists a unique $g \in \mathcal{L}^q(\Omega)$ such that the dual product*

$$\langle g^*, f \rangle_{(\mathcal{L}^p)', \mathcal{L}^p} = \int_\Omega g(x) f(x) \, \mathrm{d}x \qquad \forall f \in \mathcal{L}^p(\Omega)$$

*For details see [4, Chapter 4].*

Remark 2.6 justifies the following statements for weak and weak* convergence on $\mathcal{L}^p$-spaces, which will be used most in this text.

**Proposition 2.7** (Weak Convergence in $\mathcal{L}^p$)**.** *Let $p \in [1, \infty)$. Let $(f_n)_n \subseteq \mathcal{L}^p(\Omega)$ be a sequence and $f \in \mathcal{L}^p(\Omega)$ a function. Then*

$$f_n \rightharpoonup f \text{ in } \mathcal{L}^p(\Omega)$$

*if and only if*

$$\int_\Omega f_n(x)g(x)\,\mathrm{d}x \to \int_\Omega f(x)g(x)\,\mathrm{d}x$$

*for all* $g \in \mathcal{L}^q(\Omega)$ *with* $\frac{1}{p} + \frac{1}{q} = 1$.

*Proof.* This follows by using the Definition 2.4 of weak convergence together with the integral representation supplied by Remark 2.6. □

**Proposition 2.8** (Weak* Convergence in $\mathcal{L}^q$)**.** *Let* $q \in (1, \infty]$, $(f_n)_n \subseteq \mathcal{L}^q(\Omega)$ *be a sequence and* $f \in \mathcal{L}^q(\Omega)$ *a function. Then*

$$f_n \overset{*}{\rightharpoonup} f \text{ in } \mathcal{L}^q(\Omega)$$

*if and only if*

$$\int_\Omega f_n(x)g(x)\,\mathrm{d}x \to \int_\Omega f(x)g(x)\,\mathrm{d}x$$

*for all* $g \in \mathcal{L}^p(\Omega)$ *with* $\frac{1}{p} + \frac{1}{q} = 1$. *If it is clear from the context we write* $f_n \overset{*}{\rightharpoonup} f$ *in* $\mathcal{L}^q$.

*Proof.* This follows by using the Definition 2.5 of weak* convergence together with the integral representation supplied by Remark 2.6. □

**Theorem 2.9** (Weak* Convergence Criterion)**.** *Let* $I := (0,1) \subset \mathbb{R}$ *and let* $(f_n)_n \subseteq \mathcal{L}^\infty(I)$ *be a sequence such that* $\sup_{n\in\mathbb{N}} \|f_n\|_\infty \leq C_1 < \infty$ *and* $f \in \mathcal{L}^\infty(I)$ *such that* $\|f\|_\infty \leq C_2 < \infty$. *In order to show* $f_n \overset{*}{\rightharpoonup} f$ *in* $\mathcal{L}^\infty(I)$, *it suffices to show the following*

$$\int_a^b (f_n - f)\,dx \to 0 \qquad \forall a, b \in I, a < b \tag{2}$$

*Proof.* We want to use the following density result from [17, Chapter 7, Proposition 10]: The subspace of step functions on $[0,1]$ is dense in $\mathcal{L}^1([0,1])$, i.e., there exists a sequence $(g_l)_l$ in $\mathcal{L}^1([0,1])$ with $g_l$ defined constant on a partition of $[0,1]$ into pairwise disjoint intervals, i.e.,

$$\dot{\bigcup}_{i=1}^{N^l} (a_i^l, b_i^l) = [0,1]$$

and

$$g_l := \sum_{i=1}^{N^l} c_i^l \chi_{(a_i^l, b_i^l)}$$

for all $l \in \mathbb{N}$ such that $g_l \to g$ in $\mathcal{L}^1([0,1])$. Our interval $I = (0,1)$ in the statement is open. Since the set $\{0,1\}$ has Lebesgue measure zero, integrals over $[0,1]$ and $(0,1)$ agree for functions in $\mathcal{L}^1$, and we may use the density of step functions without loss of generality in our argument.

Let $g \in \mathcal{L}^1(I)$ be arbitrary and a sequence of step functions $(g_l)_l \subseteq \mathcal{L}^1(I)$ such that $g_l \to g$ in $\mathcal{L}^1(I)$ and assume (2) holds. To show the weak* convergence of $(f_n)_n$ we first estimate:

$$\begin{aligned}
\left|\int_0^1 g(x)(f_n(x) - f(x))\,\mathrm{d}x\right| &\leq \left|\int_0^1 (g(x) - g_l(x))(f_n(x) - f(x))\,\mathrm{d}x\right| \\
&\quad + \left|\int_0^1 g_l(x)(f_n(x) - f(x))\,\mathrm{d}x\right| \\
&\leq \|g - g_l\|_1 \|f_n - f\|_\infty + \left|\sum_{i=1}^{N^l} c_i^l \int_{a_i^l}^{b_i^l} (f_n(x) - f(x))\,\mathrm{d}x\right| \\
&\leq \|g - g_l\|_1 (C_1 + C_2) + \sum_{i=0}^{N^l} |c_i^l| \left|\int_{a_i^l}^{b_i^l} (f_n(x) - f(x))\,\mathrm{d}x\right|.
\end{aligned}$$

To conclude the convergence, let $\varepsilon > 0$ be arbitrary. By the density of step functions there exists $l_0 \in \mathbb{N}$ such that

$$\|g - g_{l_0}\|_1 < \frac{\varepsilon}{2(C_1 + C_2)}.$$

Then, for fixed $l_0$, by assumption (2), choose $n_0 \in \mathbb{N}$ large enough such that for all $n \geq n_0$

$$\sum_{i=0}^{N^{l_0}} |c_i^{l_0}| \left|\int_{a_i^{l_0}}^{b_i^{l_0}} (f_n(x) - f(x))\,\mathrm{d}x\right| < \frac{\varepsilon}{2}.$$

Therefore, for all $n \geq n_0$,

$$\left|\int_0^1 g(f_n - f)\,\mathrm{d}x\right| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary,

$$\left|\int_0^1 g(f_n - f)\,\mathrm{d}x\right| \xrightarrow{n\to\infty} 0.$$

By the arbitrariness of $g$ it follows weak* convergence as in Proposition 2.8. $\square$

**Remark 2.10.** *The proof of Theorem 3.3 does not require continuity of f, in contrary to the given reference.*

### Compactness criteria.

**Theorem 2.11** (Banach-Alaoglu)**.** *[13, Corollary A.54.]*
*Let $X$ be a separable Banach space and $(f_n)_n \subseteq X'$ be a bounded sequence, i.e., $\sup_{n\in\mathbb{N}} \|f_n\|_{X'} \leq C < \infty$. Then there exists a subsequence of $(f_n)_n$ that is weak$^*$ convergent.*

**Theorem 2.12** (Metrization)**.** *[4, Theorem 3.28]*
*Let $X$ be a separable Banach space, then the closed unit ball in the dual space $X'$ is metrizable in the weak$^*$ topology.*

### Discrete Fourier Transform

**Proposition 2.13** (Properties of the Discrete Fourier Transform)**.** *[12, Chapter 97]*
*Let the $N-$periodic sequence space $\mathbb{C}^N$ be defined as*

$$\mathbb{C}^N := \{f : \{0, \dots, N-1\} \to \mathbb{C} \mid f \text{ is } N-\text{periodic}\}$$

*Then, the following properties of this space and its Fourier basis hold*

*(i) There exists a Fourier basis $(e_m)_{m=0,\dots,N-1} \subseteq \mathbb{C}^N$ with basis vectors given by*

$$e_m(j) := \exp(\frac{2\pi i}{N} jm) \qquad \text{for } j = 0, \dots, N-1.$$

*(ii) The Fourier coeffcients $\hat{f}(j)$ of a function $f \in \mathbb{C}^N$ are given by*

$$\hat{f}(j) = N^{-1} \sum_{t=0}^{N-1} f(t) e_j(-t).$$

*(iii) The Fourier transform of a convolution is the product of the Fourier transforms*

$$\widehat{(f \star g)}(j) = \hat{f}(j)\hat{g}(j) \qquad \text{for all } f, g \in \mathbb{C}^N,$$

*where convolution of $f, g \in \mathbb{C}^N$ is defined as $(f \star g)(j) = N^{-1} \sum_{s=0}^{N-1} f(s)g(t-s)$.*

*(iv) The Plancherel Identity holds true $\sum_{t=0}^{N-1} |f(t)|^2 = N \sum_{t=0}^{N-1} |\hat{f}(t)|^2$ for all $f \in \mathbb{C}^N$.*

# 3 Gamma-Convergence

For the following let $(X, d)$ be a metric space, if not stated otherwise.

## 3.1 Lower Semicontinuity.

**Definition 3.1** (Lower Semicontinuity). *[3, Definition 1.2]*
*We say that a function $f : X \to [-\infty, \infty]$ is lower semicontinuous at a point $x \in X$, if for every sequence $x_n \to x$ we have*

$$f(x) \leq \liminf_{n\to\infty} f(x_n).$$

*We say that $f$ is lower semicontinuous on $X$ if $f$ is lower semicontinuous at each point $x \in X$.*

**Proposition 3.2** (Lower Semicontinuity of the Norm). *[4, Proposition 3.5 (iii)]*
*Let $(x_n)_n$ be a sequence in a Banach space $X$. If $x_n \rightharpoonup x \in X$, then*

$$\|x\|_X \leq \liminf_{n\to\infty} \|x_n\|_X.$$

**Theorem 3.3** (Lower Semicontinuity of Integral Functionals). *[7, Theorem 3.1]*
*Let $I := (0,1) \subset \mathbb{R}$ and let $B^1 \subset \mathbb{R}^m$ be the closed unit ball. Further, let the function $f : B^1 \to \mathbb{R}$ be a measurable function and define the functional $F : \mathcal{L}^\infty(I, B^1) \to [-\infty, \infty]$ as*

$$F(u) := \int_I f\left(u(x)\right) \,\mathrm{d}x, \qquad u \in \mathcal{L}^\infty(I, B^1).$$

*If $F$ is lower semicontinuous with respect to the weak$^*$ topology on $\mathcal{L}^\infty(I, B^1)$, then $f$ is convex, i.e., for every $a, b \in B^1$ and every $\lambda \in [0,1]$*

$$f\left(\lambda a + (1-\lambda)b\right) \leq \lambda f(a) + (1-\lambda)\, f(b).$$

*Proof.* Let $a, b \in B^1$ and $\lambda \in [0,1]$ be arbitrary. We define convex combinations

$$\bar{u}(x) := \lambda a + (1-\lambda)b \in B^1, \qquad \forall x \in I$$

and

$$\bar{f}(x) := \lambda f(a) + (1-\lambda) f(b) \in \mathbb{R} \qquad \forall x \in I.$$

This yields the convexity condition

$$f(\bar{u}(x)) \leq \bar{f}(x) \qquad \forall x \in I.$$

To show this inequality, we define the piecewise–constant functions $u_n : I \to B^1$ in $\mathcal{L}^\infty(I, B^1)$ for $n = 0, 1, 2, \ldots$ as:

$$u_0(x) := \begin{cases} a, & 0 \leq x \leq \lambda, \\ b, & \lambda < x \leq 1, \end{cases}$$

and

$$u_n(x) := u_0(nx - \lfloor nx \rfloor) \qquad n \geq 1$$

for all $x \in I$. We are done if we show that $u_n \overset{*}{\rightharpoonup} \bar{u}$ and $f(u_n) \overset{*}{\rightharpoonup} \bar{f}$ in $\mathcal{L}^\infty$. We justify this: By assumption $F$ is lower semicontinuous. In particular for the test function $\chi_{[0,1]} \in \mathcal{L}^1(I, B^1)$ the weak* convergence holds, so

$$f(\bar{u}(x)) = F(\bar{u}) \leq \liminf_{n\to\infty} F(u_n) = \liminf_{n\to\infty} \int_0^1 f(u_n(x))dx = \bar{f}(x)$$

for all $x \in I$ and thus $f$ is convex.

We claim that $u_n \overset{*}{\rightharpoonup} \bar{u}$ and $f(u_n) \overset{*}{\rightharpoonup} \bar{f}$ in $\mathcal{L}^\infty$. We have $|u_n| \leq \max\{|a|, |b|\} < \infty$ and $|f(u_n)| \leq \max\{|f(a)|, |f(b)|\} < \infty$ for all $n = 0, 1, 2, \ldots$. Therefore, by Theorem 2.9, it suffices to show that

$$\int_s^t g_n(x)\,\mathrm{d}x \to \int_s^t g(x)\,\mathrm{d}x \qquad \forall s, t \in I, s < t$$

in order to show $g_n \overset{*}{\rightharpoonup} g$ in $\mathcal{L}^\infty$. To show $u_n \overset{*}{\rightharpoonup} \bar{u}$ in $\mathcal{L}^\infty$, let $s, t \in I, s < t$ be arbitrary. Then

$$\begin{aligned}
\left|\int_s^t (\bar{u}(x) - u_n(x))dx\right| &= \left|\int_s^t (\bar{u}(x) - u_0(nx - \lfloor nx \rfloor)dx\right| \\
&= \left|\int_s^t \left(\bar{u}(x) - a\chi_{[0,\lambda]}(nx - \lfloor nx \rfloor) - b\chi_{(\lambda,1]}(nx - \lfloor nx \rfloor)\right) dx\right| \\
&\leq \left|\left(t - s - \frac{1}{n}(\lfloor tn \rfloor - \lceil sn \rceil)\right)(\lambda a + (1-\lambda)b)) + \frac{2}{n}(a+b)\right| \\
&\leq \left|\left(t - \frac{\lfloor tn \rfloor}{n} + \frac{\lceil sn \rceil}{n} - s\right)(\lambda a + (1-\lambda)b)) + \frac{2}{n}(a+b)\right| \\
&\overset{n\to\infty}{\longrightarrow} 0.
\end{aligned}$$

Similarly for $f(u_n)$: let $s, t \in I, s < t$ be arbitrary. Then

$$\begin{aligned}
&\left|\int_s^t \left(\bar{f}(x) - f(u_n(x))\right) dx\right| \\
=&\left|\int_s^t \left(\bar{f}(x) f(u_0(nx - \lfloor nx \rfloor))\right) dx\right| \\
=&\left|\int_s^t \left(\bar{f}(x) - f(a)\chi_{[0,\lambda]}(nx - \lfloor nx \rfloor) - f(b)\chi_{(\lambda,1]}(nx - \lfloor nx \rfloor)\right) dx\right| \\
\leq&\left|\left(t - \frac{\lfloor tn \rfloor}{n} + \frac{\lceil sn \rceil}{n} - s\right)(\lambda f(a) + (1-\lambda)f(b)) + \frac{2}{n}(f(a) + f(b))\right| \\
&\overset{n\to\infty}{\longrightarrow} 0.
\end{aligned}$$

□

**Remark 3.4** (Metrization on $\mathcal{L}^\infty$)**.** *Indeed $\mathcal{L}^\infty(I, B^1)$ equipped with the weak$^*$ topology is metrizable. It is $\mathcal{L}^1(I, \mathbb{R}^m)$ a Banach space by the Fischer-Riesz Theorem [4, Theorem 4.8] and $\mathcal{L}^1(I, \mathbb{R}^m)$ is separable by [4, Theorem 4.13]. As a consequence of Theorem 2.12, the closed unit ball of the dual space is metrizable in the weak$^*$ topology.*

## 3.2 General Theory of Gamma-Convergence

**Definition 3.5** ($\Gamma-$Convergence)**.** *[3, Definition 1.5]*
*We say that a sequence of functions $E_n : X \to [-\infty, \infty]$ for $n \in \mathbb{N}$ $\Gamma-$converges in $X$ to $E : X \to [-\infty, \infty]$ if for all $x \in X$*

*(i)* $\liminf$ *inequality*

$$\forall x_n \to x : E(x) \leq \liminf_{n\to\infty} E_n(x_n)$$

*(ii)* $\limsup$ *inequality*

$$\exists x_n \to x : E(x) \geq \limsup_{n\to\infty} E_n(x_n).$$

*We say $E$ is the $\Gamma-$limit of $(E_n)_n$ and write $E = \Gamma - \lim_n E_n$. The sequence $(x_n)_n$ for the* $\limsup$ *inequality is called recovery sequence.*

**Remark 3.6.** *In the above definition of $\Gamma-$convergence we deduce, that the* $\limsup$ *inequality is in fact an equality.*

*Proof.* By the $\liminf$ and $\limsup$ inequality we have for a recovery sequence $x_n \to x$ that $E(x) \leq \liminf_{n\to\infty} E_n(x_n) \leq \limsup_{n\to\infty} E_n(x_n) \leq E(x)$. This implies that the $\limsup$ inequality may be replaced by

$$\exists x_n \to x : E(x) = \lim_{n\to\infty} E_n(x_n).$$

□

To give the reader a first intuition on $\Gamma-$convergence, we provide an example. The pointwise and $\Gamma-$limits may exist, but be different at every point.

**Example 3.7.** *Consider $E_n : \mathbb{R} \to \mathbb{R}$ for $n \in \mathbb{N}$ given by*

$$E_n(x) := \begin{cases} 0 & \text{if } x \notin \mathbb{Q} \text{ or } x = \frac{k}{m}, k \in \mathbb{Z}, m \in \{1, \dots, n\} \\ -1 & \text{otherwise.} \end{cases}$$

*Then $E_n \to 0$ pointwise, but $\Gamma - \lim_{n\to\infty} E_n \equiv -1$.*

*Proof.* Let $x \in \mathbb{R}$ be arbitrary. For any sequence $x_n \to x$ we have

$$-1 \leq \liminf_{n\to\infty} E_n(x_n).$$

Note that the preimage

$$M_n := \{y \in \mathbb{R} \mid E_n(y) = -1\} = \mathbb{Q}\backslash\{\frac{k}{m} \mid m \leq n, k \in \mathbb{Z}\}$$

is dense in $\mathbb{R}$ of for each $n \in \mathbb{N}$. Therefore there exists a sequence $(y_n)_n$ with $y_n \in M_n$ for all $n \in \mathbb{N}$, such that $y_n \to x$ and $\limsup_{n\to\infty} E_n(y_n) \equiv -1$. Since $x \in \mathbb{R}$ was arbitrary, it follows that $\Gamma - \lim_{n\to\infty} E_n \equiv -1$ as claimed. □

We will state two useful properties of $\Gamma-$convergence.

**Lemma 3.8** (Lower Semicontinuity of the Γ-limit)**.** *[3, Proposition 1.28]*
*Let $E_n : X \to [-\infty, \infty]$ be a sequence of functions and $E : X \to [-\infty, \infty]$ a function. If $\Gamma - \lim_n E_n = E$, then $E$ is lower semicontinuous.*

*Proof.* Follows from the indicated reference and Remark 1.25 in [3] pointing out the definition used in this text. □

**Remark 3.9.** *In Lemma 3.8, we make use of the fact that $X$ is a metric space.*

In the following we state and prove a key theorem in the theory of $\Gamma-$convergence under the following coercivity assumption.

**Definition 3.10** (Equi-Coercivity)**.** *Let $E_n : X \to [-\infty, \infty]$ be a sequence of functions. The sequence $(E_n)_n$ is equi-coercive, if for every sequence $(x_n)_n$ in $X$ there holds:*

$$\sup_{n\in\mathbb{N}} E_n(x_n) \leq C < \infty \Longrightarrow (x_n)_n \text{ has a convergent subsequence.}$$

**Remark 3.11.** *In Definition 1.19 in [3] mild-equi-coercivity is introduced. In Theorem 3.12 below, unlike in the reference, we assume equi-coercivity instead of mild-equi-coercivity, which is stronger.*

**Theorem 3.12** (Fundamental Theorem of Γ-convergence)**.** *[3, Theorem 1.21]*
*Let $E_n : X \to [-\infty, \infty]$ be a sequence of functions and $E : X \to [-\infty, \infty]$ a function such that*

*(A1)* $(E_n)_n$ *$\Gamma-$converges to $E$*

*(A2)* $(E_n)_n$ *is equi-coercive and*

*(A3)* $(\inf_X E_n)_n$ *is bounded.*

*Then*

$$\min_X E = \lim_{n\to\infty} \left(\inf_X E_n\right) \tag{3}$$

*and $\bar{x} \in X$ is minimizer of $E$ if and only if $\bar{x}$ is an accumulation point of some $(x_n)_n$ such that*

$$\lim_{n\to\infty} \left| E_n(x_n) - \inf_X E_n \right| = 0, \tag{4}$$

*i.e., there exists a subsequence $(x_{n_k})_k \subseteq (x_n)_n$ such that $x_{n_k} \to \bar{x}$.*

For the proof we use the following result.

**Lemma 3.13** (Subsubsequence Criterion)**.** *Let $(x_n)_n \subseteq X$ be a sequence and $x \in \mathbb{R}$. If every subsequence $(x_{n_k})_k \subseteq (x_n)_n$ has a convergent subsubsequence $(x_{n_{k_j}})_j \subseteq (x_{n_k})_k$ such that $x_{n_{k_j}} \to x$, then $x_n \to x$.*

*Proof.* Let $x \in \mathbb{R}$. For the sake of contradiction suppose that $(x_n)_n$ does not converge to $x$, i.e

$$\exists \varepsilon > 0 \, \forall l \in \mathbb{N} \, \exists n_l \geq l : |x_{n_l} - x| > \varepsilon.$$

This defines a subsequence $(x_{n_l})_l \subseteq (x_n)_n$. For every $(x_{n_{l_m}})_m \subseteq (x_{n_l})_l$ then holds

$$\forall l \in \mathbb{N} \, \exists m \geq l : |x_{n_{l_m}} - x| > \varepsilon.$$

In particular, there exists no $N \in \mathbb{N}$ such that for all $j \geq N$ we have $|x_{n_{l_j}} - x| < \varepsilon$, i.e., there exists no convergent subsubsequence in contradiction to the assumption. □

*Proof of Theorem 3.12.* Assume conditions (A1), (A2), (A3) are satisfied.

**Step 1** We show if $\bar{x}$ is an accumulation point of some $(x_n)_n$ such that

$$\lim_{n\to\infty} \left| E_n(x_n) - \inf_X E_n \right| = 0,$$

then

$$\min_X E = \lim_{n\to\infty} \left(\inf_X E_n\right)$$

and $\bar{x} \in X$ is a minimizer.

Thereto, we consider an asymptotically minimizing sequence $(x_n)_n$ in $X$ such that (4) holds true, i.e., $\lim_{n\to\infty} |E_n(x_n) - \inf_X E_n| = 0$. By assumption (A3) the sequence $(\inf_X E_n)_n$ is bounded, i.e., there exists $K \in \mathbb{R}$ such that $\inf_X E_n \leq K$ for all $n \in \mathbb{N}$. Then there exists $M \in \mathbb{R}$ such that $E_n(x_n) \leq M$ for all $n \in \mathbb{N}$ for the minimizing sequence $(x_n)_n$.

We are now able to use the equi-coercivity condition (A2) and obtain a convergent subsequence $x_{n_k} \to \bar{x} \in X$ of $(x_n)_n$.

We will now show, that indeed $\bar{x}$ is a minimizer of $E$. From the lim inf inequality we get

$$E(\bar{x}) \leq \liminf_{k\to\infty} E_{n_k}(x_{n_k}) \overset{(4)}{=} \liminf_{k\to\infty} \left(\inf_X E_{n_k}\right). \tag{5}$$

Let now $y \in X$ be arbitrary. The lim sup inequality provides for a recovery sequence $y_n \to y$. In particular for the subsequence $(n_k)_k$ above in (5) there holds $y_{n_k} \to y$. Now with Remark 3.6 for the first equality, we deduce

$$\begin{aligned} E(y) &= \limsup_{n\to\infty} E_n(y_n) \\ &\geq \limsup_{k\to\infty} E_{n_k}(y_{n_k}) \\ &\geq \limsup_{k\to\infty} \left(\inf_X E_{n_k}\right) \\ &\geq \liminf_{k\to\infty} \left(\inf_X E_{n_k}\right) \\ &\overset{(5)}{\geq} E(\bar{x}). \end{aligned}$$

This shows that $\bar{x}$ is a minimizer of $E$.

To show (3), i.e., $\min_X E = \lim_{n\to\infty} (\inf_X E_n)$, we use the Subsubsequence Criterion (Lemma 3.13). Consider any subsequence $(\inf_X E_{n_j})_j$ of $(\inf_X E_n)_n$, then by equi-coercivity (A2) we get a convergent subsequence $x_{n_{j_l}} \to x^*$. Perform the same computation from (5) until here, it follows by the same arguments. Finally pick $y = x^*$ and the inequalities become equalities, yielding the minimum value

$$E(x^*) = \lim_{l\to\infty} (\inf_X E_{n_{j_l}}).$$

Thus every subsubsequence converges to the same limit $\min_X E$ and with Lemma 3.13, that leads to

$$\min_{x\in X} E(x) = \lim_{n\to\infty} \left(\inf_{x\in X} E_n(x)\right),$$

which proves (3).

**Step 2** To prove the reverse direction let, $\tilde{x} \in X$ be a minimzer of $E$. By the lim sup inequality there exists a recovery sequence $\tilde{x}_n \to \tilde{x}$ such that:

$$\begin{aligned} \lim_{n\to\infty} E_n(\tilde{x}_n) &= E(\tilde{x}) \\ &= \min_{x\in X} E(x) \\ &\overset{(3)}{=} \lim_{n\to\infty} \left(\inf_X E_n\right), \end{aligned}$$

implying $\lim_{n\to\infty} |E_n(\tilde{x}_n) - \inf_{x\in X} E_n(x)| = 0$, which is exactly (4). □

## 3.3 Application to Discrete Spin Systems

Having established the general theory of Γ-convergence, we present this abstract result, which will be employed to compute the $\Gamma-$limit of the energy functional in

the next section.

**Theorem 3.14.** *[1, Theorem 3.3]*
*Let $K \subseteq \mathbb{R}^m$ be a bounded set. For all $\xi \in \mathbb{Z}$ let $f^\xi : \mathbb{R}^{2m} \to \mathbb{R}$ be a function such that:*

*(H1)* $\forall \xi \in \mathbb{Z} : f^\xi(u,v) = f^{-\xi}(v,u)$,

*(H2)* $\forall \xi \in \mathbb{Z} : f^\xi(u,v) = +\infty$ *if* $(u,v) \notin K^2$,

*(H3) for all $\xi$, there exists $C_\xi \geq 0$ such that $|f^\xi(u,v)| \leq C_\xi$ for all $(u,v) \in K^2$, and $\sum_\xi C_\xi < \infty$.*

*Let $\lambda_n$ be a vanishing sequence of positive numbers. Let $I := (0,1) \subseteq \mathbb{R}$ and define the set of discrete points*

$$\mathbb{Z}_n(I) := \{i \in \mathbb{Z} : \lambda_n i \in \bar{I}\}.$$

*Furthermore, we denote by $\lambda_n(i + [0,1)) := [\lambda_n i, \lambda_n(i+1))$ and define the set of piecewise constant functions as*

$$\mathcal{D}_n(I, \mathbb{R}^m) := \{u : \mathbb{R} \to \mathbb{R}^m \mid u \text{ constant on } \lambda_n(i + [0,1))] \text{ for all } i \in \mathbb{Z}_n(I)\}.$$

*Let $F_n : \mathcal{L}^\infty(I, \mathbb{R}^m) \to (-\infty, +\infty]$ be defined by*

$$F_n(u) = \begin{cases} \sum_{\xi \in \mathbb{Z}} \sum_{i \in R_n^\xi(I)} \lambda_n f^\xi(u(i), u(i+\xi)) & \text{if } u \in \mathcal{D}_n(I, \mathbb{R}^m), \\ +\infty & \text{otherwise,} \end{cases}$$

*where $R_n^\xi(I) := \{i \in \mathbb{Z}_n(I) : i + \xi \in \mathbb{Z}_n(I)\}$.*

*Then $(F_n)_n$ $\Gamma$-converges with respect to the weak* topology in $\mathcal{L}^\infty(I)$ to $F : \mathcal{L}^\infty(I, \mathbb{R}^2) \to (-\infty, +\infty]$*

$$F(u) = \begin{cases} \int_I f_{\text{hom}}(u(x))\,\mathrm{d}x & \text{if } u \in \mathcal{L}^\infty(I, \mathrm{co}(K)) \\ +\infty & \text{otherwise.} \end{cases} \tag{6}$$

*Here, $\mathrm{co}(K)$ denotes the convex hull of $K$ and the homogenized energy density $f_{\text{hom}}$ is given by*

$$f_{\text{hom}}(z) := \lim_{\rho \to 0} \lim_{k \to \infty} \frac{1}{k} \inf \Bigg\{ \sum_{\xi \in \mathbb{Z}} \sum_{\beta \in R_1^\xi((0,k))} f^\xi(v(\beta), v(\beta+\xi)) \\ \mid \langle v \rangle_{1,(0,k)} \in \bar{B}(z,\rho) \Bigg\}. \tag{7}$$

*The discrete average of $v$ in $A$ is defined as*

$$\langle v \rangle_{1,A} := \frac{1}{\#(\mathbb{Z} \cap A)} \sum_{i \in \mathbb{Z} \cap A} v(i).$$

*Proof.* Can be found in the given reference. □

# 4 Analysis of the Energy Model

In this section we present the $\Gamma-$convergence of the energy functionals $(E_n)_n$ and we characterize minimizers and minima of $E_n$ as well as of the $\Gamma-$limit $E$, scaled in $\alpha$. Then, we analyze the helimagnetic/ferromagnetic transition point. In the last part, we take a look at an energy model of higher order.

We denote by $S^1 \subset \mathbb{R}^2$ the unit sphere and by $B^1 \subset \mathbb{R}^2$ the closed unit ball. Throughout Section 4 we fix a vanishing sequence of positive numbers $(\lambda_n)_n$, i.e. $\lambda_n > 0$ for all $n \in \mathbb{N}$ with $\lambda_n \to 0$ and set $I := (0,1) \subset \mathbb{R}$.

Consider a discrete system on the lattice $\mathbb{Z}_n(I) := \{i \in \mathbb{Z} : \lambda_n i \in \bar{I}\}$. The set of state variables is denoted by

$$\mathcal{U}_n(I, S^1) := \{u : i \in \mathbb{Z}_n(I) \mapsto u^i \in S^1\}.$$

Such a frustrated spin system is driven by an energy $E_n : \mathcal{U}_n(I, S^1) \to \mathbb{R}$ given as

$$E_n(u) = -J_1 \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + J_2 \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}),$$

for interaction parameters $J_1 > 0$, $J_2 > 0$. Setting the interaction ratio $\alpha := J_1/J_2$, we will consider the family of energies

$$E_n(u) = -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}),$$

where $E_n$ refers to $E_n/J_2$. Further, we fix periodic boundary conditions on the system to avoid edge artifacts:

$$(u^0, u^1) = (u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor}). \tag{8}$$

For those state variables satisfying the boundary conditions, we define the set

$$\bar{\mathcal{U}}_n(I, S^1) := \{u : \mathbb{Z}_n(I) \to S^1 | u \text{ satisfies } (u^0, u^1) = (u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor})\}.$$

In order to analyse asymptotic behavior of the discrete systems, we embed the family of functionals on a conjoint functional space, by extending $E_n$ to the Lebesgue

space $\mathcal{L}^\infty(I, B^1)$. To this end, we associate to any $u \in \bar{\mathcal{U}}_n(I, S^1)$ a piecewise constant interpolation belonging to the class

$$\mathcal{C}_n(I, S^1) := \{u \in \bar{\mathcal{U}}_n(I, S^1) | u(x) = u(\lambda_n i) \text{ if } x \in \lambda_n(i + [0, 1)), \text{ for all } i \in \mathbb{Z}_n(I)\},$$

which is a subspace of $\mathcal{L}^\infty(I, B^1)$. We keep the notation for $u \in \bar{\mathcal{U}}_n(I, S^1)$ such that $u^i := u(\lambda_n i)$. As a consequence we will from now on consider the extension of the energies on $\mathcal{L}^\infty(I, B^1)$ via $E_n : \mathcal{L}^\infty(I, B^1) \to (-\infty, \infty]$

$$E_n(u) = \begin{cases} -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}) & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise.} \end{cases} \tag{9}$$

For the definitions of the state-variable sets introduced in Sections 3 and 4, see the notational summary in Section 2. We will use those sets repeatedly in the proofs below.

## 4.1 Characterization of Ground States

It is adivsable to reformulate $E_n$ in terms of a non-negative functional $H_n$ such that

$$E_n(u) = H_n(u) - (1 + \frac{\alpha^2}{8})(1 - c_n \lambda_n). \tag{10}$$

Now, minimizing $E_n$ reduces to finding $u$ such that $H_n(u) = 0$. This simplifies the proof of Proposition 4.1.

We define the functional $H_n : \mathcal{L}^\infty(I, B^1) \to (-\infty, \infty]$ as

$$H_n(u) = \begin{cases} \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left| u^{i+2} - \frac{\alpha}{2} u^{i+1} + u^i \right|^2 & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise.} \end{cases} \tag{11}$$

Dissolving the quadratic summand and noticing that $|u^i|^2 = 1$ for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor - 2$, we obtain

$$\begin{aligned} &\left| u^{i+2} - \frac{\alpha}{2} u^{i+1} + u^i \right|^2 \\ &= \left| u^{i+2} \right|^2 + \frac{\alpha^2}{4} |u^{i+1}|^2 + |u^i|^2 - \alpha(u^{i+2}, u^{i+1}) + 2(u^{i+2}, u^i) - \alpha(u^{i+1}, u^i) \\ &= 2 + \frac{\alpha^2}{4} + 2(u^{i+2}, u^i) - \alpha \left( (u^{i+2}, u^{i+1}) + (u^{i+1}, u^i) \right). \end{aligned}$$

In order to unite the NN terms, execute an index shift $j = i+1$ and since $u \in \bar{\mathcal{U}}_n(I, S^1)$

the periodic boundary condition $(u^0, u^1) = (u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor})$ holds, we get

$$\sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^{i+2}, u^{i+1}) = \sum_{j=1}^{\lfloor 1/\lambda_n \rfloor - 1} \lambda_n(u^{j+1}, u^j) = \sum_{j=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^{j+1}, u^j).$$

Rename the index $j = i$ once again. Then define $c_n := \frac{1}{\lambda_n} - \left\lfloor \frac{1}{\lambda_n} \right\rfloor + 1 \in [1, 2)$, so that

$$\sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n = 1 - 1 + \lambda_n(\lfloor 1/\lambda_n \rfloor - 1) = 1 - c_n \lambda_n \xrightarrow{n \to \infty} 1. \tag{12}$$

Gathering things together we can write $H_n$ as

$$\begin{aligned} H_n(u) &= \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left(1 + \frac{\alpha^2}{8}\right) - \alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}) \\ &= (1 - c_n \lambda_n) \left(1 + \frac{\alpha^2}{8}\right) - E_n(u), \end{aligned}$$

as intended in (10).

In the next proposition we characterize global minimizers of $(E_n)_n$.

**Proposition 4.1** (Characterization of minimizers of $E_n$)**.** *[5, Proposition 3.2] Let $0 < \alpha \leq 4$. Let $E_n : \mathcal{L}^\infty(I, B^1) \to (-\infty, \infty]$ be given by*

$$E_n(u) = \begin{cases} -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}) & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise} \end{cases}$$

*for $n \in \mathbb{N}$, as in (9) with $\mathcal{C}_n(I, S^1)$ as in (1). Then*

$$\min_{\mathcal{L}^\infty(I, B^1)} E_n = -\left(1 + \frac{\alpha^2}{8}\right)(1 - c_n \lambda_n), \tag{13}$$

*where $c_n := \frac{1}{\lambda_n} - \left\lfloor \frac{1}{\lambda_n} \right\rfloor + 1$. Furthermore, a minimizer $u_n$ of $E_n$ over $\mathcal{L}^\infty(I, B^1)$ for $n \in \mathbb{N}$ satisfies*

$$(u_n^i, u_n^{i+1}) = \frac{\alpha}{4} \text{ and } (u_n^i, u_n^{i+2}) = \frac{\alpha^2}{8} - 1 \tag{14}$$

*for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor - 2$.*

*Proof.* Let $0 < \alpha \leq 4$ and let $E_n$ be as in (9).

**Step 1** We show the minima of $(E_n)_n$ in (13).

Let $H_n$ be defined as in (11). Since $H_n \geq 0$ by (10) we deduce

$$E_n(u) = H_n(u) - (1 + \frac{\alpha^2}{8})(1 - c_n\lambda_n) \geq -\left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n)$$

for all $u \in \mathcal{L}^\infty(I, B^1)$ and all $n \in \mathbb{N}$. In order to show, the minimum is attained, we construct a minimizing sequence for $(E_n)_n$, that is, a sequence $(u_n)_n$ with $H_n(u_n) = 0$ for all $n \in \mathbb{N}$. Fix $\varphi \in [0, \pi]$ such that

$$\cos\varphi = \frac{\alpha}{4}. \tag{15}$$

Then we construct $(u_n)_n$ by setting $u_n : \mathbb{Z}_n \to S^1$ as

$$u_n^i := \begin{pmatrix} \cos(\varphi i) \\ \sin(\varphi i) \end{pmatrix} \tag{16}$$

for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor$, for all $n \in \mathbb{N}$.

We show $u_n \in \mathcal{C}_n(I, S^1)$ (1) for every $n \in \mathbb{N}$. By the Pythagorean trigonometric identity $|u_n^i|^2 = 1$ for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor$ and all $n \in \mathbb{N}$. Further, the sequence $(u_n)_n$ is defined piecewise constant on lattice cells. With help of the angle subtraction theorem for cos-function, we show angles inbetween spins are fixed by design

$$\begin{aligned} (u_n^{i+1}, u_n^i) &= \cos(\varphi(i+1))\cos(\varphi i) + \sin(\varphi(i+1))\sin(\varphi i) \\ &= \cos(\varphi(i+1) - \varphi i) \\ &= \cos(\varphi) \\ &\overset{(15)}{=} \frac{\alpha}{4} \end{aligned} \tag{17}$$

and thus $u_n$ satisfies the periodic boundary condition (8) for every $n \in \mathbb{N}$. Therefore we conclude that $u_n \in \mathcal{C}_n(I, S^1)$ for every $n \in \mathbb{N}$ indeed. We will now establish that $(u_n)_n$ is a minimizing sequence for $(E_n)_n$ by showing $H_n(u_n) = 0$ for all $n \in \mathbb{N}$. We compute

$$\begin{aligned} u_n^i + u_n^{i+2} &= \begin{pmatrix} \cos(\varphi i) + \cos(\varphi(i+2)) \\ \sin(\varphi i) + \sin(\varphi(i+2)) \end{pmatrix} \\ &= 2\begin{pmatrix} \cos\left(\frac{\varphi i + \varphi(i+2)}{2}\right)\cos\left(\frac{\varphi i - \varphi(i+2)}{2}\right) \\ \sin\left(\frac{\varphi i + \varphi(i+2)}{2}\right)\cos\left(\frac{\varphi i - \varphi(i+2)}{2}\right) \end{pmatrix} \\ &= 2\begin{pmatrix} \cos(\varphi(i+1))\cos(\varphi) \\ \sin(\varphi(i+1))\cos(\varphi) \end{pmatrix} \\ &= 2\cos(\varphi)u_n^{i+1} \\ &\overset{(15)}{=} \frac{\alpha}{2}u_n^{i+1} \end{aligned}$$

for all $i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 2$, for all $n \in \mathbb{N}$ . Here, we used trigonometric identities derived from the angle addition and subtraction theorems as follows

$$\begin{aligned} &\cos(a+b) + \cos(a-b) = 2\cos(a)\cos(b) \\ &a := \frac{x+y}{2}, b := \frac{x-y}{2} \\ \Longrightarrow &\cos(x) + \cos(y) = 2\cos(\frac{x+y}{2})\cos(\frac{x-y}{2}) \end{aligned}$$

and similarly for sin-function:

$$\begin{aligned} &\sin(a+b) + \sin(a-b) = 2\sin(a)\cos(b) \\ &a := \frac{x+y}{2}, b := \frac{x-y}{2} \\ \Longrightarrow &\sin(x) + \sin(y) = 2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2}). \end{aligned}$$

The result $u_n^i + u_n^{i+2} = \frac{\alpha}{2} u_n^{i+1}$ implies $H_n(u_n) = 0$ and thus

$$E_n(u_n) = -\left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n)$$

for every $n \in \mathbb{N}$ and (13) is proven.

**Step 2** We show the form of the minimizers in (14).

Let's consider a minimizer $\tilde{u}_n$ of $E_n$ over $\mathcal{L}^\infty(I, B^1)$ for $n \in \mathbb{N}$. By definition of $E_n$ in (9), we have that $\tilde{u}_n \in \mathcal{C}_n(I, S^1)$. By the characterization of minima in (13) and by the relation of $E_n$ and $H_n$ in (10) it must be $H_n(\tilde{u}_n) = 0$, which implies

$$\tilde{u}_n^{i+2} - \frac{\alpha}{2}\tilde{u}_n^{i+1} + \tilde{u}_n^i = 0 \Longleftrightarrow \tilde{u}_n^{i+1} = \frac{2}{\alpha}(\tilde{u}_n^i + \tilde{u}_n^{i+2}) \tag{18}$$

for all $i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 2$, for all $n \in \mathbb{N}$. Using that $\tilde{u}_n$ is $S^1-$valued we compute

$$1 = \left|\tilde{u}_n^{i+1}\right|^2 = \frac{4}{\alpha^2}\left|\tilde{u}_n^i + \tilde{u}_n^{i+2}\right|^2 = \frac{8}{\alpha^2}\left(1 + (\tilde{u}_n^i, \tilde{u}_n^{i+2})\right) \tag{19}$$

and

$$(\tilde{u}_n^i, \tilde{u}_n^{i+1}) \overset{(18)}{=} \frac{2}{\alpha}(\tilde{u}_n^i, \tilde{u}_n^i + \tilde{u}_n^{i+2}) = \frac{2}{\alpha}(1 + \left(\tilde{u}_n^i, \tilde{u}_n^{i+2}\right)) \overset{(19)}{=} \frac{\alpha}{4}$$

for all $i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 2$, for all $n \in \mathbb{N}$, which shows the claim (14). □

**Remark 4.2.** *Examining the case $\alpha \geq 4$. The ground states are all ferromagnetic, i.e., for any $u \in \bar{\mathcal{U}}_n(I, S^1)$ we have $u^i = \bar{u}$ for all $i = 0, \dots, \lfloor 1/\lambda_n \rfloor$ and some $\bar{u} \in S^1$.*

*Proof.* To show this, let $E_n^4$ be the energy for $\alpha = 4$ and compute for $E_n$ for some

$\alpha \geq 4$:

$$E_n(u) = -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n (u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n (u^i, u^{i+2})$$
$$= E_n^4(u) - (\alpha - 4) \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n (u^i, u^{i+1}).$$

By Proposition 4.1 the minimizing states of $E_n^4$ satisfy (14):

$$(u_n^i, u_n^{i+1}) = 1 \text{ and } (u_n^i, u_n^{i+2}) = 1$$

and are hence ferromagnetic states. The expression above is minimized if the second term is maximized in absolute values:

$$\left| \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n (u^i, u^{i+1}) \right| \leq \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n |(u^i, u^{i+1})| \leq \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n |u^i||u^{i+1}| \leq \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n,$$

which holds true for ferromagnetic states. Therefore we get the minimum value for each $n \in \mathbb{N}$ as follows.

$$\begin{aligned}
\min E_n(u) &= \min E_n^4 - (\alpha - 4) \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \\
&\overset{(13)}{=} -\left(1 + \frac{4^2}{8}\right)(1 - c_n \lambda_n) - (\alpha - 4) \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \\
&\overset{(12)}{=} -\left(1 + \frac{4^2}{8}\right)(1 - c_n \lambda_n) - (\alpha - 4)(1 - c_n \lambda_n) \\
&= -(\alpha - 1)(1 - c_n \lambda_n).
\end{aligned}$$

□

In the following example, we reproduce the result from Proposition 4.1 for $n = 5$ by minimizing explicitly and provide visualization in Fig. 2 of the minimizing state.

**Example 4.3.** *Set up a lattice with cell length $\lambda_n := \frac{1}{n}$ for every $n \in \mathbb{N}$, so that $(\lambda_n)_n$ is a zero sequence. Fix an angle $\varphi \in [0, \pi]$ and assume $0 < \alpha \leq 4$. For instance, consider $n = 5$, where the lattice points are*

$$\mathbb{Z}_5(I) = \{0, \ldots, \lfloor 1/\lambda_5 \rfloor\} = \{0, \ldots, 5\}$$

*and a helical state $u_5$ defined by*

$$u_5^i := \begin{pmatrix} \cos \varphi i \\ \sin \varphi i \end{pmatrix}$$

*for all $i \in \mathbb{Z}_5$. This is a special case of the minimizing sequence (16) from the proof of Proposition 4.1 for $n = 5$. Therefore, we have $u_5 \in \mathcal{C}_5(I, S^1)$. We compute the energy as in (9).*

$$\begin{aligned} E_5(u_5) &= -\alpha \sum_{i=0}^{3} \frac{1}{5}(u_5^i, u_5^{i+1}) + \sum_{i=0}^{3} \frac{1}{5}(u_5^i, u_5^{i+2}) \\ &= -\alpha \sum_{i=0}^{3} \frac{1}{5} \cos(\varphi) + \sum_{i=0}^{3} \frac{1}{5} \cos(2\varphi) \\ &= -\alpha \frac{4}{5} \cos(\varphi) + \frac{4}{5} \cos(2\varphi) \\ &= -\alpha \frac{4}{5} \cos(\varphi) + \frac{4}{5}(2\cos^2(\varphi) - 1). \end{aligned}$$

*We now minimze with respect to $\cos(\varphi)$:*

$$0 \overset{!}{=} \frac{dE_5(u)}{d\cos(\varphi)} = -\alpha \frac{4}{5} + \frac{16}{5} \cos(\varphi).$$

*Then*

$$(u_5^i, u_5^{i+1}) = \cos(\varphi) = \frac{\alpha}{4} \text{ and } (u_5^i, u_5^{i+2}) = 2\cos(\varphi)^2 - 1 = \frac{\alpha^2}{8} - 1,$$

*as expected. This matches the result (14) in Proposition 4.1. With help of the minimizer $u_5$ for the angle $\varphi_{\min} := \arccos(\alpha/4)$ we compute the minimal value for the energy*

$$\begin{aligned} E_5(u_5) &= -\alpha \frac{4}{5} \frac{\alpha}{4} + \frac{4}{5} \left( \frac{\alpha^2}{8} - 1 \right) \\ &= -\frac{\alpha^2}{10} - \frac{4}{5} \\ &= -\left( \frac{\alpha^2}{8} + 1 \right) \left( 1 - \frac{1}{5} \right) \\ &= -\left( \frac{\alpha^2}{8} + 1 \right) \left( 1 - \left( \frac{1}{5} \right) (5 - \lfloor 1/5 \rfloor + 1) \right) \\ &= -\left( \frac{\alpha^2}{8} + 1 \right) (1 - \lambda_5 c_5) \end{aligned}$$

*in accordance with (13).*

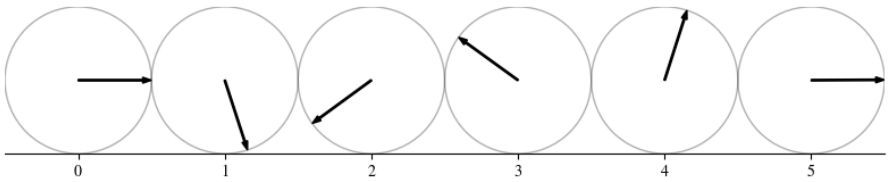


Figure 2: Optimal minimizing spin configuration with constant angle between adjacent spins. Counter-clockwise spin rotation. Interaction ratio $\alpha = 1.24$. Lattice size $n = 5$.

## 4.2 The Gamma-Limit of the Energy Functional

In the following we present the $\Gamma-$Convergence of the energy functionals $(E_n)_n$ and we characterize minimizers and minima of the $\Gamma-$limit $E$, scaled in $\alpha$.

For the following, we equip $\mathcal{L}^\infty(I, B^1)$ with the weak* topology. By Remark 3.4 it is metrizable and in particular we may use the results from Section 3.

**Theorem 4.4** ($\Gamma-$Convergence of the Energy)**.** *[5, Theorem 3.4]*
*Let $E_n : \mathcal{L}^\infty(I, B^1) \to (-\infty, \infty]$ for $n \in \mathbb{N}$ be given by*

$$E_n(u) = \begin{cases} -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}) & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise,} \end{cases}$$

*as in (9) with $\mathcal{C}_n(I, S^1)$ as in (1). Then $(E_n)_n$ $\Gamma-$converges with respect to the weak* on $\mathcal{L}^\infty(I, B^1)$ and the $\Gamma-$limit is given by*

$$E(u) = \int_I f_{\text{hom}}(u(x))dx \qquad \text{for all } u \in \mathcal{L}^\infty(I, B^1), \tag{20}$$

*where the function $f_{\text{hom}} : B^1 \to \mathbb{R}$ is given by the following asymptotic homogenization formula:*

$$f_{\text{hom}}(z) := \lim_{\rho \to 0} \lim_{k \to \infty} \frac{1}{k} \inf \left\{ -\alpha \sum_{i=0}^{k-2} (u^i, u^{i+1}) + \sum_{i=0}^{k-2} (u^i, u^{i+2}) \mid \langle u \rangle_{1,(0,k)} \in \bar{B}(z, \rho) \right\}$$

*for all $z \in B^1$.*

*Proof.* We use an abstract result Theorem 3.14 to obtain the convergence.

We first show that the sequence of auxiliary functionals $(F_n)_n$ from Theorem 3.14 $\Gamma-$converges to $F$. Then, with help of $(F_n)_n$, we show lim inf and lim sup inequalities such that $(E_n)_n$ $\Gamma-$converges to $E$. This intermediate step is needed, since $(E_n)_n$ is only defined on $\mathcal{L}^\infty(I, B^1) \subset \mathcal{L}^\infty(I, \mathbb{R}^2)$ and $(F_n)_n$ is defined on all of $\mathcal{L}^\infty(I, \mathbb{R}^2)$.

**Step 1** We prove $\Gamma - \lim_n F_n = F$ by using Theorem 3.14.

Here, $F_n : \mathcal{L}^\infty(I, \mathbb{R}^2) \to (-\infty, +\infty]$ is given by

$$F_n(u) = \begin{cases} \sum\limits_{\xi\in\mathbb{Z}} \sum\limits_{i\in R_n^\xi(I)} \lambda_n f^\xi(u^i, u^{i+\xi}) & \text{if } u \in \mathcal{D}_n(I, \mathbb{R}^2), \\ +\infty & \text{otherwise,} \end{cases} \tag{21}$$

where $\mathcal{D}_n(I, \mathbb{R}^2)$ as in (1). For given $\alpha > 0$, we define the function $f^\xi : \mathbb{R}^4 \to \mathbb{R}$ for $(u, v) \in K^2 := S^1 \times S^1$ and $\xi \in \mathbb{Z}$ as:

$$f^\xi(u, v) = \begin{cases} -\frac{\alpha}{2}(u, v) & \text{if } |\xi| = 1, (u, v) \in K^2, \\ \frac{1}{2}(u, v) & \text{if } |\xi| = 2, (u, v) \in K^2, \\ 0 & \text{if } |\xi| \notin \{1, 2\}, (u, v) \in K^2 \\ +\infty & \text{if } (u, v) \notin K^2. \end{cases} \tag{22}$$

We verify the assumptions for $f^\xi$ in Theorem 3.14 given by

(H1) $\forall \xi \in \mathbb{Z} : f^\xi(u, v) = f^{-\xi}(v, u)$,

(H2) $\forall \xi \in \mathbb{Z} : f^\xi(u, v) = +\infty$ if $(u, v) \notin K^2$,

(H3) for all $\xi$, there exists $C_\xi \geq 0$ such that $|f^\xi(u, v)| \leq C_\xi$ for all $(u, v) \in K^2$, and $\sum\limits_\xi C_\xi < \infty$.

The assumption (H1) is satisfied, since $f^\xi$ is defined symmetrically in $\xi$ via the absolute value function and the inner product is symmetric. The second assumption (H2) is satisfied by the definition of $f^\xi$. The third assumption (H3) follows with the Cauchy-Schwarz inequality, since for any $(u, v) \in K^2$

$$\begin{aligned} \left|f^\xi(u, v)\right| &\leq \frac{1}{2}\max\{1, \alpha\}|u||v| = \frac{1}{2}\max\{1, \alpha\} =: C_\xi < \infty, \\ \sum_\xi C_\xi &\leq 2\max\{1, \alpha\} < +\infty. \end{aligned}$$

Notice that the convex hull $co(S^1) = B^1$. Then, by Theorem 3.14 (6) we have the $\Gamma-$limit of $(F_n)_n$ with respect to the weak* topology on $\mathcal{L}^\infty$ given by $F : \mathcal{L}^\infty(I, \mathbb{R}^2) \to$

$(-\infty, +\infty]$ as

$$F(u) = \begin{cases} \int_I f_{\text{hom}}(u(x))\,\mathrm{d}x & \text{if } u \in \mathcal{L}^\infty(I, B^1) \\ +\infty & \text{otherwise,} \end{cases}$$

where $f_{\text{hom}} : B^1 \to \mathbb{R}$ is given by

$$f_{\text{hom}}(z) = \lim_{\rho \to 0} \lim_{k \to \infty} \frac{1}{k} \inf \left\{ \sum_{\xi \in \mathbb{Z}} \sum_{\beta \in \mathcal{R}_1^\xi((0,k))} f^\xi(v^\beta, v^{\beta+\xi}) \mid \langle v \rangle_{1,(0,k)} \in \bar{B}(z, \rho) \right\}.$$

Note, that $E(u) = F(u)$ for all $u \in \mathcal{L}^\infty(I, B^1)$. To see that $f_{\text{hom}}$ is indeed as in the definition of $E$ in (20), we plug in the definition of $f^\xi$ and compute the sums.

$$\begin{aligned} \sum_{\xi \in \mathbb{Z}} \sum_{i \in R_n^\xi(I)} f^\xi(u^i, u^{i+\xi}) &= \sum_{i=0}^{k-2} \frac{-\alpha}{2}(u^i, u^{i+1}) + \sum_{i=2}^{k} \frac{-\alpha}{2}(u^i, u^{i-1}) \\ &\quad + \sum_{i=0}^{k-2} \frac{1}{2}(u^i, u^{i+2}) + \sum_{i=2}^{k} \frac{1}{2}(u^i, u^{i-2}) \\ &= -\alpha \sum_{i=0}^{k-2} (u^i, u^{i+1}) + \sum_{i=0}^{k-2} (u^i, u^{i+2}). \end{aligned} \tag{23}$$

For the last equality an index shift was carried out and for the NN terms we used the periodic boundary condition from (8) given by

$$(u^0, u^1) = (u^{k-1}, u^k).$$

This demonstrates the homogenization formula of $f_{\text{hom}}$.

**Step 2** We show the lim inf inequality for $(E_n)_n$.

In the following computation we observe that the functionals $E_n$ and $F_n$ coincide

on $\mathcal{C}_n(I,S^1)$ (1) for every $n \in \mathbb{N}$. For any $u \in \mathcal{C}_n(I,S^1)$ we have

$$\begin{aligned}
F_n(u) &= \sum_{\xi\in\mathbb{Z}} \sum_{i\in R_n^\xi(I)} \lambda_n f^\xi(u^i, u^{i+\xi}) \\
&= \sum_{i=0}^{\lfloor 1/\lambda_n\rfloor -2} \lambda_n \frac{-\alpha}{2}(u^i, u^{i+1}) + \sum_{i=2}^{\lfloor 1/\lambda_n\rfloor} \lambda_n \frac{-\alpha}{2}(u^i, u^{i-1}) \\
&+ \sum_{i=0}^{\lfloor 1/\lambda_n\rfloor -2} \lambda_n \frac{1}{2}(u^i, u^{i+2}) + \sum_{i=2}^{\lfloor 1/\lambda_n\rfloor} \lambda_n \frac{1}{2}(u^i, u^{i-2}) \\
&= -\alpha \sum_{i=0}^{\lfloor 1/\lambda_n\rfloor -2} \lambda_n (u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n\rfloor -2} \lambda_n (u^i, u^{i+2}) \\
&= E_n(u).
\end{aligned}$$

for every $n \in \mathbb{N}$, where in the last step an index shift was carried out and we used the periodic boundary condition as in (23). Furthermore, by definition of $E_n$ in (9) we have $E_n(u) = +\infty$ for all other $u \in \mathcal{L}^\infty(I,B^1)$. Therefore, we have

$$E_n(u) \geq F_n(u) \text{ for all } u \in \mathcal{L}^\infty(I,B^1).$$

Thus, we obtain

$$\forall u_n \overset{*}{\rightharpoonup} u \text{ in } \mathcal{L}^\infty(I,B^1) : \liminf_{n\to\infty} E_n(u_n) \geq \liminf_{n\to\infty} F_n(u_n) = E(u),$$

which is the lim inf inequality from Definition 3.5.

**Step 3** We show the lim sup inequality for $(E_n)_n$.

If $v \in \mathcal{L}^\infty(I,B^1)$ such that $E(v) = +\infty$, there is nothing to show. Therefore, consider $v \in \mathcal{L}^\infty(I,B^1)$ with $E(v) < +\infty$. We have $E(u) = F(u)$ for all $u \in \mathcal{L}^\infty(I,B^1)$ by Step 2. Then, by definition of $F_n$ in (21) for this specific $f^\xi$ and by Definition 3.5 ($\Gamma-$Convergence) there exists a recovery sequence $v_n \in \mathcal{D}_n(I,S^1)$ for every $n \in \mathbb{N}$ such that $v_n \overset{*}{\rightharpoonup} v$ in $\mathcal{L}^\infty(I,B^1)$ and by Remark 3.6 $E(v) = \lim_{n\to\infty} F_n(v_n)$. The set $\mathcal{D}_n(I,S^1)$ (1) in contrary to $\mathcal{C}_n(I,S^1)$ (1) does not include periodic boundary conditions. The recovery sequence may violate the periodic boundary condition. To avoid this, we define a sequence $w_n \in \mathcal{C}_n(I,S^1)$ for every $n \in \mathbb{N}$ by rotating a single spin

$$w_n^i := \begin{cases} v_n^i & \text{if } i \neq 1 \\ R_n v_n^1 & \text{if } i = 1, \end{cases}$$

where $R_n \in SO(2)$ is such that

$$(v_n^0, R_n v_n^1) = (v_n^{\lfloor 1/\lambda_n\rfloor -1}, v_n^{\lfloor 1/\lambda_n\rfloor}).$$

We claim that $w_n \overset{*}{\rightharpoonup} v$ in $\mathcal{L}^\infty(I, B^1)$, because $(w_n)_n$ and $(v_n)_n$ only differ on one lattice cell. To prove this, we use the Weak* Convergence Criterion Theorem 2.9. Since $\|w_n\|_\infty = \|v_n\|_\infty \leq 1 < \infty$ and $\|v\|_\infty \leq 1 < \infty$, it suffices to show

$$\int_a^b (w_n - v)dx \to 0 \qquad \forall a, b \in I, a < b$$

To this end let $a, b \in I, a < b$ be arbitrary. Then

$$\left| \int_a^b (w_n(x) - v_n(x))\,\mathrm{d}x \right| \leq \lambda_n \big|(R_n v_n^1 - v_n^1)\big| \leq \lambda_n 2 \overset{n\to\infty}{\longrightarrow} 0,$$

and thus for all $a, b \in I, a < b$:

$$\int_a^b (w_n(x) - v(x))\,\mathrm{d}x = \int_a^b (w_n(x) - v_n(x))\,\mathrm{d}x + \int_a^b (v_n(x) - v(x))\,\mathrm{d}x \to 0.$$

This shows that $w_n \overset{*}{\rightharpoonup} v$ in $\mathcal{L}^\infty(I, B^1)$.

Now, we compute the difference of $E_n(w_n)$ and $F_n(v_n)$:

$$\begin{aligned}
&|E_n(w_n) - F_n(v_n)| \\
&= \Bigg| \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(-\alpha(w_n^i, w_n^{i+1}) + (w_n^i, w_n^{i+2})) \\
&- \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \frac{-\alpha}{2}(v^i, v^{i+1}) - \sum_{i=1}^{\lfloor 1/\lambda_n \rfloor - 1} \lambda_n \frac{-\alpha}{2}(v^{i+1}, v^i) - \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n (v^i, v^{i+2}) \Bigg| \\
&\overset{p.b.c.}{=} \lambda_n | - \alpha(w_n^0, w_n^1) - \alpha(w_n^1, w_n^2) + (w_n^1, w_n^3) \\
&+ \frac{\alpha}{2}(v^0, v^1) + \alpha(v^1, v^2) - (v^1, v^3) + \frac{\alpha}{2}(, v^{N-1}, v^N)| \\
&\leq \lambda_n(4\alpha + 2) \overset{n\to\infty}{\longrightarrow} 0.
\end{aligned}$$

Here, we have used the periodic boundary conditions (8). Therefore,

$$\lim_{n\to\infty} E_n(w_n) = \lim_{n\to\infty} F_n(v_n) = E(v),$$

which is the lim sup inequality as in Remark 3.6. Together with Step 2 we obtain

$$\Gamma - \lim_n E_n = E,$$

as claimed in (20). □

**Remark 4.5** (Implicit Extension of the Functionals)**.** *Throughout this thesis we define the functionals on $\mathcal{L}^\infty(I, B^1)$, because the weak$^*$ topology is metrizable on $\mathcal{L}^\infty(I, B^1)$. By contrast, the original paper [5] extends the functionals, at times only implicitly, to $\mathcal{L}^\infty(I, \mathbb{R}^2)$, which is not a metric space with respect to the same topology.*

*The exposition of $\Gamma$-convergence in Section 3 is formulated for metric spaces. Nevertheless, the theory extends to arbitrary topological spaces, in particular Definition 3.5 can be formulated in the sense of topological convergence (see [6, Chapter 4]).*

*Consequently, the $\Gamma$-convergence result proved here agrees with that of [5].*

**Example 4.6.** *Fig. 3 displays the minimal energy of the $\Gamma-$limit $E(u)$. It shows the energy in the helimagnetic regime continuously transitioning into the linear energy profile of the ferromagnetic regime.*

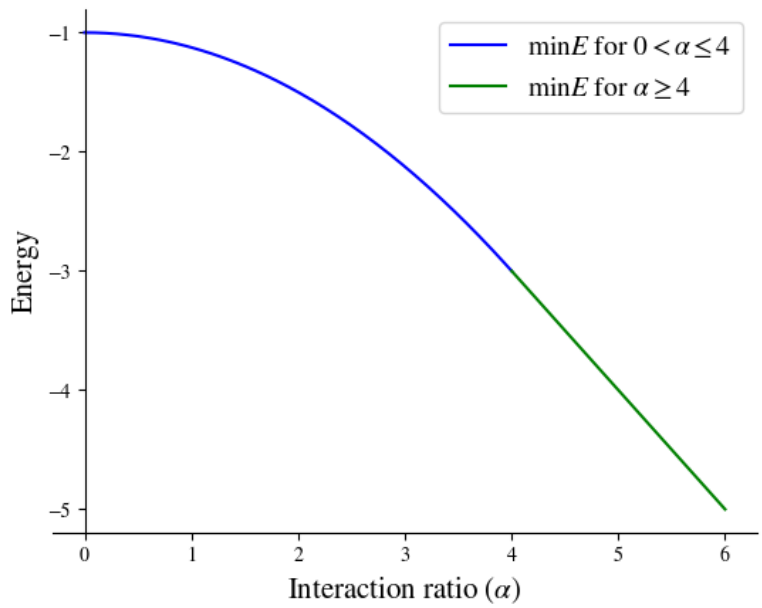


Figure 3: The minimal energy of the $\Gamma$-limit, $\min E(u)$, as a function of the interaction ratio $\alpha$. The plot illustrates the phase transition from the helimagnetic ground state ($0 < \alpha \leq 4$) to the ferromagnetic ground state ($\alpha \geq 4$). The minimal energy at the transition point ($\alpha = 4$) is continuous, reflecting the shift in the system's energetic ground state as described in Theorem 4.4

**Theorem 4.7** (Characterization of Minimizers of $E$)**.** *[5, Theorem 3.4]*
*Let $E : \mathcal{L}^\infty(I, B^1) \to (-\infty, +\infty]$ be given by*

$$E(u) = \int_I f_{\text{hom}}(u(x))dx \qquad \textit{for all } u \in \mathcal{L}^\infty(I, B^1)$$

*where* $f_{\text{hom}} : B^1 \to \mathbb{R}$ *is*

$$f_{\text{hom}}(z) := \lim_{\rho\to 0}\lim_{k\to\infty}\frac{1}{k}\inf\left\{-\alpha\sum_{i=0}^{k-2}(u^i,u^{i+1}) + \sum_{i=0}^{k-2}(u^i,u^{i+2}) \mid \langle u\rangle_{1,(0,k)} \in \bar{B}(z,\rho)\right\}$$

*for all* $z \in B^1$*. Then,*

*(i)* $f_{\text{hom}}$ *is convex on* $B^1$*,*

*(ii) if* $0 < \alpha \leq 4$*, then the following estimate holds:*

$$\frac{(\alpha-4)^2}{8}|z|^2 \leq f_{\text{hom}}(z) + \left(1+\frac{\alpha^2}{8}\right) \leq \frac{(\alpha-4)^2}{8}|z|$$

*for all* $z \in B^1$*,*

*(iii) there exists* $h : [0,1] \to \mathbb{R}$ *which is convex and non-decreasing such that* $f_{\text{hom}}(z) = h(|z|)$ *for all* $z \in B^1$*.*

*(iv) if* $0 < \alpha \leq 4$*, we have that*

$$\min_{\mathcal{L}^\infty(I,B^1)} E = E(0) = -\left(1+\frac{\alpha^2}{8}\right),$$

*and* $u \equiv 0$ *is the unique minimizer in* $\mathcal{L}^\infty(I,B^1)$*,*

*(v) if* $\alpha \geq 4$*, then*

$$\min_{\mathcal{L}^\infty(I,B^1)} E = E(u) = -(\alpha-1) \qquad \text{for all } u \in \mathcal{L}^\infty(I,B^1),$$

*since* $f_{\text{hom}}(z) = -(\alpha-1)$ *for all* $z \in B^1$*.*

*Proof.* The key step, identifying $E$ as the $\Gamma-$limit of $(E_n)_n$, has been proven in Theorem 4.4. This allows us to use the Fundamental Theorem of $\Gamma-$ convergence (Theorem 3.12), which ensures the convergence of minimizers and minima. We will proceed as follows. In a first step we verify the assumptions of the Fundamental Theorem of $\Gamma-$Convergence, i.e., we recall that the sequence $(E_n)_n$ $\Gamma-$converges to $E$ given by an integral function with the energy density $f_{\text{hom}}$, we show that the sequence $(E_n)_n$ is equi-coercive and that the sequence $(\inf E_n)_n$ is bounded. In a second step, we characterize the energy density $f_{\text{hom}}$ explicitly, including the derivation of lower and upper bounds and further properties necessary for the final part of the proof. In the third step, we compute the minimal value of $E$ and describe its minimizers in the helimagnetic ($0 < \alpha \leq 4$) and ferromagnetic ($\alpha \geq 4$) regime.

## Step 1

We justify the application of the Fundamental Theorem of $\Gamma-$Convergence 3.12. Let $(E_n)_n$ be defined as in (9) and let us verify the following assumptions.

(1) $(E_n)_n$ $\Gamma-$converges to $E$

(2) $(E_n)_n$ is equi-coercive and

(3) $(\inf_{\mathcal{L}^\infty(I,B^1)} E_n)_n$ is bounded.

### Step 1.1

By Theorem 4.4 we have $(E_n)_n$ $\Gamma-$converges to $E$.

### Step 1.2

We show the sequence $(E_n)_n$ is equi-coercive, i.e., for every sequence $(u_n)_n$ in $\mathcal{L}^\infty(I, B^1)$ there holds:

$$\sup_{n\in\mathbb{N}} E_n(u_n) \leq C < \infty \Longrightarrow (x_n)_n \text{ has a convergent subsequence,}$$

see Definition 3.10.

To obtain this let $(u_n)_n$ be a sequence in $\mathcal{L}^\infty(I, B^1)$ and assume that

$$\sup_{n\in\mathbb{N}} E_n(u_n) \leq C < \infty.$$

Then, by definition of $(E_n)_n$ in (9), we have $u_n \in \mathcal{C}_n(I, S^1)$ for every $n \in \mathbb{N}$, so that $\sup_{n\in\mathbb{N}} \|u_n\|_\infty \leq 1$. Since $\mathcal{L}^1$ is separable Banach space and $\mathcal{L}^\infty$ its dual, we may now use Banach-Alaoglu (Theorem 2.11) and get a weak$^*$ convergent subsequence of $(u_n)_n$.

### Step 1.3

We verify $(\inf_{\mathcal{L}^\infty(I,B^1)} E_n)_n$ is bounded.

First, let $0 < \alpha \leq 4$. In Proposition 4.1 we have shown

$$\inf_{\mathcal{L}^\infty(I,B^1)} E_n = \min_{\mathcal{L}^\infty(I,B^1)} E_n = -\left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n)$$

for all $n \in \mathbb{N}$. Second, let $\alpha \geq 4$. In Remark 4.2 we have shown

$$\inf_{\mathcal{L}^\infty(I,B^1)} E_n = \min_{\mathcal{L}^\infty(I,B^1)} E_n = -(\alpha - 1)(1 - c_n\lambda_n)$$

for all $n \in \mathbb{N}$. Further by (12) we have $c_n\lambda_n \overset{n\to\infty}{\longrightarrow} 0$. Therefore the sequence of infima is a sequence of minima and is a bounded sequence.

We may now apply the Fundamental Theorem of $\Gamma-$Convergence 3.12 in the subsequent steps of the proof.

## Step 2

In Step 1.1 we have established the $\Gamma-$limit $E$ and its energy density $f_{\text{hom}}$. Now, we show some properties of $f_{\text{hom}}$, which are needed to characterize the minimal values of $E$ and its minimizers.

### Step 2.1

We employ Lower Semicontinuity of Integral Functionals (Theorem 3.3) to obtain the convexity of $f_{\text{hom}}$. In Step 1.1 we identified $E$ as the $\Gamma-$limit of $(E_n)$. By Lower Semicontinuity of the $\Gamma-$limit (Lemma 3.8) this implies that $E$ is lower semicontinuous with respect to the weak* topology on $\mathcal{L}^\infty(I, B^1)$. Then choose $K = B^1$ as the convex set in Theorem 3.3 and $f = f_{\text{hom}}$. The homogenization formula defines $f_{\text{hom}}$ as a pointwise limit of infima of continuous functions in $z$. Hence $f_{\text{hom}}$ is measurable on $B^1$. All assumptions are therefore satisfied, so we conclude (i) convexity of $f_{\text{hom}}$ on $B^1$.

### Step 2.2

We prove a lower bound for $f_{\text{hom}}$ for $0 < \alpha \leq 4$, that is

$$\frac{(\alpha-4)^2}{8}|z|^2 \leq f_{\text{hom}}(z) + (1 + \frac{\alpha^2}{8}) \text{ for all } z \in B^1. \tag{24}$$

To show this, let $\alpha \in (0, 4]$.

**Step 2.2.1** We claim that we are done if we presume

$$\liminf_{n\to\infty} H_n(u_n) \geq \frac{(\alpha-4)^2}{8} \int_I |u(x)|^2 \, \mathrm{d}x \tag{25}$$

for all sequences $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty(I, B^1)$ with $u_n \in \mathcal{C}_n(I, S^1)$ (1) for every $n \in \mathbb{N}$.

To justify this, we use the relation between $H_n$ and $E_n$ from (10). Assume (25) holds true, then

$$\begin{aligned}\frac{(\alpha-4)^2}{8} \int_I |u(x)|^2 \, dx &\leq \liminf_{n\to\infty} \left( E_n(u_n) + (1 + \frac{\alpha^2}{8})(1 - c_n\lambda_n) \right) \\ &\overset{(12)}{=} \liminf_{n\to\infty} E_n(u_n) + (1 + \frac{\alpha^2}{8})\end{aligned}$$

for all $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$. By Theorem 4.4 $(E_n)_n$ $\Gamma-$converges to $E$. Therefore, the computation is in particular correct for the recovery sequence $(v_n)_n \in \mathcal{L}^\infty(I, B^1)$ that converges weakly* to $v \in \mathcal{L}^\infty(I, B^1)$, where $v(x) := z \in B^1$ is constant. As pointed out in Remark 3.6 the lim sup inequality is in fact an equality, which implies

$$\begin{aligned}
\frac{(\alpha-4)^2}{8}|z|^2 &\leq \lim_{n\to\infty} E_n(v_n) + (1+\frac{\alpha^2}{8}) \\
&= E(v) + (1+\frac{\alpha^2}{8}) \\
&= \int_I f_{\text{hom}}(z)\,\mathrm{d}x + \left(1+\frac{\alpha^2}{8}\right) \\
&= f_{\text{hom}}(z) + \left(1+\frac{\alpha^2}{8}\right),
\end{aligned}$$

which is the lower bound in (24).

**Step 2.2.2** We show

$$\liminf_{n\to\infty} H_n(u_n) \geq \frac{(\alpha-4)^2}{8} \int_I |u(x)|^2\,\mathrm{d}x$$

for all sequences $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty(I, B^1)$ with $u_n \in \mathcal{C}_n(I, S^1)$ for every $n \in \mathbb{N}$.

Let $u_n \in \mathcal{C}_n(I, S^1)$ for $n \in \mathbb{N}$ such that $u_n \overset{*}{\rightharpoonup} u \in \mathcal{L}^\infty(I, B^1)$. We claim that there exist functions $w_n, \tilde{u}_n \in \mathcal{C}_n(I, S^1)$ with $w_n \overset{*}{\rightharpoonup} u$ and $\tilde{u}_n \overset{*}{\rightharpoonup} u$ such that we have

$$H_n(u_n) \geq 2\int_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x - \lambda_n\frac{\alpha^2}{8}. \tag{26}$$

In Step 2.2.3 we prove such functions exist. In addition, we use weak lower semicontinuity of the $\mathcal{L}^2-$norm from Proposition 3.2 to obtain the estimate. Notice that for all $n \in \mathbb{N}$ the functions $w_n, \hat{u}_n, u_n$ and $u$ are in $\mathcal{L}^\infty(I)$. Since the domain has finite measure $|I| = 1 < \infty$ those function are in $\mathcal{L}^2(I)$ as well. Here the exemplary computation for $u_n$:

$$\|u_n\|_2^2 = \int_I |u_n(x)|^2\,\mathrm{d}x \leq \|u_n\|_\infty^2 \leq 1 < \infty.$$

Furthermore, for domains with finite measure the inclusion $\mathcal{L}^2(I) \subset \mathcal{L}^1(I)$ holds true. Test functions for weak* convergence in $\mathcal{L}^\infty(I)$ are in $\mathcal{L}^1(I)$, therefore testing with $\mathcal{L}^2(I)$-functions is valid and $w_n, \hat{u}_n$ converge weakly to $u$ in $\mathcal{L}^2(I)$ as in Proposition 2.7.

We conclude the lower bound (ii) with lower semicontinuity of the $\mathcal{L}^2-$norm:

$$\begin{aligned}\liminf_{n\to\infty} H_n(u_n) &\overset{(26)}{\geq} \liminf_{n\to\infty} 2\int_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x \\ &\overset{\text{l.s.c.}}{\geq} 2\int_I \left|u(x) - \frac{\alpha}{4}u(x)\right|^2 \mathrm{d}x \\ &= \frac{(\alpha-4)^2}{8}\int_I |u(x)|^2 \mathrm{d}x.\end{aligned}$$

**Step 2.2.3** We show that there exist functions $w_n, \tilde{u}_n \in \mathcal{C}_n(I, S^1)$ with $w_n \overset{*}{\rightharpoonup} u$ and $\tilde{u}_n \overset{*}{\rightharpoonup} u$ such that we have

$$H_n(u_n) \geq 2\int_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x - \lambda_n \frac{\alpha^2}{8}.$$

To prove this, let $u_n \in \mathcal{C}_n(I, S^1)$ for $n \in \mathbb{N}$ such that $u_n \overset{*}{\rightharpoonup} u \in \mathcal{L}^\infty(I, B^1)$. We define $w_n \in \mathcal{C}_n(I, S^1)$ for $n \in \mathbb{N}$ as

$$w_n^i = \begin{cases} \frac{u_n^i + u_n^{i+2}}{2} & \text{if } i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 2 \\ 0 & \text{if } i = \lfloor 1/\lambda_n \rfloor - 1, \end{cases}$$

and we define the functions $\hat{u}_n \in \mathcal{C}_n(I, S^1)$ for $n \in \mathbb{N}$ as

$$\hat{u}_n \coloneqq u_n^{i+1} \text{ for } i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 1.$$

Then $\sup_{n\in\mathbb{N}} \|w_n\|_\infty \leq \sup_{n\in\mathbb{N}} \|u_n\|_\infty$ and $\sup_{n\in\mathbb{N}} \|\hat{u}_n\|_\infty \leq \sup_{n\in\mathbb{N}} \|u_n\|$. Now, we

write $H_n(u_n)$ in terms of $w_n$ and $\hat{u}_n$ as follows:

$$\begin{aligned}
H_n(u_n) &= \frac{1}{2}\sum_{i=0}^{\lfloor 1/\lambda_n\rfloor-2} \lambda_n\left|u_n^{i+2} - \frac{\alpha}{2}u_n^{i+1} + u_n^i\right|^2 \\
&= 2\sum_{i=0}^{\lfloor 1/\lambda_n\rfloor-2} \lambda_n\left|\frac{u_n^{i+1}+u_n^{i+2}}{2} - \frac{\alpha}{4}u_n^i\right|^2 \\
&= 2\sum_{i=0}^{\lfloor 1/\lambda_n\rfloor-2} \int\limits_{\lambda_n[i,i+1)} \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x \\
&= 2\int\limits_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x - 2\lambda_n\left|w_n^{\lfloor 1/\lambda_n\rfloor-1} - \frac{\alpha}{4}\hat{u}_n^{\lfloor 1/\lambda_n\rfloor-1}\right|^2 \\
&= 2\int\limits_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x - 2\lambda_n\left|\frac{\alpha}{4}u_n^{\lfloor 1/\lambda_n\rfloor}\right|^2 \\
&\geq 2\int\limits_I \left|w_n(x) - \frac{\alpha}{4}\hat{u}_n(x)\right|^2 \mathrm{d}x - \lambda_n\frac{\alpha^2}{8}.
\end{aligned}$$

We still have to show that $w_n \overset{*}{\rightharpoonup} u$ and $\hat{u}_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$. We start with $w_n$. By Theorem 2.9 this is equivalent to showing that

$$\int\limits_a^b w_n(x)\,\mathrm{d}x \to \int\limits_a^b u(x)\,\mathrm{d}x \qquad \forall a,b \in I, a<b$$

By this computation

$$\int\limits_a^b (w_n(x) - u(x))\,\mathrm{d}x = \int\limits_a^b (w_n(x) - u_n(x))\,\mathrm{d}x + \int\limits_a^b (u_n(x) - u(x))\,\mathrm{d}x$$

and the fact that $u_n \overset{*}{\rightharpoonup} u$, we justify that it suffices to show

$$\int\limits_a^b (u_n(x) - w_n(x))\,\mathrm{d}x \to 0 \qquad \forall a,b \in I, a<b.$$

To show this, let $a,b \in I, a<b$ and set $a_n := \lfloor a/\lambda_n\rfloor$ and $b_n := \lfloor b/\lambda_n\rfloor$. Then, we rewrite the integral by breaking it into a sum over the discrete intervals $[\lambda_n i, \lambda_n(i+1))$

and accounting for the boundaries at $a$ and $b$.

$$\begin{aligned}
&\int_a^b (w_n(x) - u_n(x))\,\mathrm{d}x \\
&= \sum_{i=a_n}^{b_n} \int_{\lambda_n[i,i+1)} (u_n^i - w_n^i)\,\mathrm{d}x - \int_{[a_n,a]} (u_n^{a_n} - w_n^{a_n})\,\mathrm{d}x - \int_{[b,b_n+1)} (u_n^{b_n} - w_n^{b_n})\,\mathrm{d}x \\
&= \sum_{i=a_n}^{b_n} \lambda_n(u_n^i - w_n^i) + \underbrace{(a-a_n)}_{\leq\lambda_n}(u_n^{a_n} - w_n^{a_n}) + \underbrace{(b_n+1-b)}_{\leq\lambda_n}(u_n^{b_n} - w_n^{b_n}). \qquad (27)
\end{aligned}$$

Now, we focus on the term $|w_n^i - u_n^i|$. For $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor - 2$ write the difference as:

$$u_n^i - w_n^i = u_n^i - \frac{u_n^i + u_n^{i+2}}{2} = \frac{1}{2}(u_n^i - u_n^{i+2}).$$

In case that $i = \lfloor 1/\lambda_n \rfloor - 1$, we obtain $u_n^i - w_n^i = u_n^i$. This yields a telescopic sum, which can be condensed as follows:

$$\sum_{i=a_n}^{b_n} (u_n^i - w_n^i) = \begin{cases} \frac{1}{2}\left(u_n^{a_n} + u_n^{a_n+1} - u_n^{b_n+1} - u_n^{b_n+2}\right) & \text{if } b_n < \lfloor 1/\lambda_n \rfloor - 1 \\ \frac{1}{2}\left(u_n^{a_n} + u_n^{a_n+1} + u_n^{\lfloor 1/\lambda_n \rfloor - 1} - u_n^{\lfloor 1/\lambda_n \rfloor}\right) & \text{if } b_n = \lfloor 1/\lambda_n \rfloor - 1. \end{cases}$$

Then, we estimate the telescopic sum from above by $2\|u_n\|_\infty$ and the boundary terms each by $\|u_n\|_\infty$. Altogether,

$$\left| \int_a^b (w_n(x) - u_n(x)\,\mathrm{d}x \right| \leq \lambda_n \|u_n\|_\infty (2+1+1) \leq 4\lambda_n \overset{n\to\infty}{\longrightarrow} 0.$$

We show $\hat{u} \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$. By the same argumentation as for $w_n$, it suffices to show that

$$\int_a^b (\hat{u}_n(x) - u_n(x))\,\mathrm{d}x \to 0 \qquad \forall a, b \in I, a < b.$$

We write the integral as a discrete sum and account for the boundaries, as in (27). The telescopic sum is

$$\sum_{i=a_n}^{b_n} (u_n^i - \hat{u}^i) = u_n^{a_n} - u_n^{b_n+1},$$

and thus we can estimate it by $2\|u_n\|_\infty$ as well as the boundary terms. Now, the

convergence follows with

$$\begin{aligned}\left|\int_a^b (\hat{u}_n(x) - u_n(x)\,\mathrm{d}x\right| &\leq \lambda_n\left|u_n^{a_n} - u_n^{b_n+1}\right| + \lambda_n\left|u_n^{a_n} - u_n^{a_n+1}\right| + \lambda_n\left|u_n^{b_n} - u_n^{b_n+1}\right| \\ &\leq \lambda_n\|u_n\|_\infty(2+2+2) \\ &\leq 6\lambda_n \overset{n\to\infty}{\longrightarrow} 0.\end{aligned}$$

This finishes Step 2.2, which proved the lower bound of $f_{\text{hom}}$.

**Step 2.3**

We prove an upper bound for $f_{\text{hom}}$ for $0 < \alpha \leq 4$, that is

$$f_{\text{hom}}(z) + \left(1 + \frac{\alpha^2}{8}\right) \leq \frac{(\alpha-4)^2}{8}|z| \text{ for all } z \in B^1.$$

To show this, let $\alpha(0,4]$.

**Step 2.3.1** First, we prove

$$f_{\text{hom}}(0) = -\left(1 + \frac{\alpha^2}{8}\right). \tag{28}$$

In Step 2.2 we have established the lower bound (24), therefore we only need to show

$$f_{\text{hom}}(0) \leq -\left(1 + \frac{\alpha^2}{8}\right).$$

To this end, we construct the sequence $u_n \in \mathcal{C}_n(I, S^1)$ for every $n \in \mathbb{N}$ as in Proposition 4.1 (16):

$$u_n^i := \begin{pmatrix}\cos(\varphi i)\\ \sin(\varphi i)\end{pmatrix} \text{ for } i = 0, \ldots, \lfloor 1/\lambda_n \rfloor.$$

We show that $u_n \overset{*}{\rightharpoonup} 0$ in $\mathcal{L}^\infty$. By Theorem 2.9 it suffices to show

$$\int_a^b u_n(x)dx \to 0 \qquad \forall a, b \in I, a < b.$$

Let $a, b \in I, a < b$ be arbitrary. The periodicity of the trigonometric functions

translates to $q := \frac{2\pi}{\varphi}$ for $u_n$, i.e., $u_n^i = u_n^{i+\frac{2\pi}{\varphi}}$. Summing over one period we obtain

$$\sum_{j=0}^{q-1} u_n^{i+j} = 0 \text{ for all } i = 0, \dots, \lfloor 1/\lambda_n \rfloor.$$

Hence, for the sum over $i = \lfloor a/\lambda_n \rfloor, \dots, \lfloor b/\lambda_n \rfloor$ we only have to consider the boundary parts. The number $r$ of the leftover summands can be estimated by the period length $r \leq 2q$. Altogether,

$$\left| \int_a^b u_n(x)\,\mathrm{d}x \right| \leq \lambda_n \left| \sum_{i=\lfloor a/\lambda_n \rfloor}^{\lfloor b/\lambda_n \rfloor} u_n^i \right| + \lambda_n 2 \|u_n\|_\infty \leq \lambda_n (r+2) \|u_n\|_\infty \overset{n\to\infty}{\longrightarrow} 0.$$

This shows $u_n \overset{*}{\rightharpoonup} 0$ in $\mathcal{L}^\infty$ . Recall $(u_n)_n$ are minimizers of $(E_n)_n$ according to Proposition 4.1, which implies

$$E_n(u_n) = \min_{\mathcal{L}^\infty(I,B^1)} E_n = -\left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n) \qquad \forall n \in \mathbb{N}.$$

As $(E_n)_n$ $\Gamma-$converges to $E$ by Theorem 4.4, we can make use of the lim inf inequality for $u_n \overset{*}{\rightharpoonup} 0$ in $\mathcal{L}^\infty$ as follows:

$$f_{\text{hom}}(0) = \int_I f_{\text{hom}}(0)\,\mathrm{d}x = E(0) \leq \liminf_{n\to\infty} E_n(u_n) = -\left(1 + \frac{\alpha^2}{8}\right). \tag{29}$$

This yields $f_{\text{hom}}(0) = -\left(1 + \frac{\alpha^2}{8}\right)$.

**Step 2.3.2** We prove the upper bound on $S^1$, i.e.,

$$f_{\text{hom}}(z) + \left(1 + \frac{\alpha^2}{8}\right) \leq \frac{(\alpha-4)^2}{8}|z| \text{ for all } z \in S^1. \tag{30}$$

To this end, let $z \in S^1$. Let us consider a constant sequence $u_n \in \mathcal{C}_n(I, S^1)$ for every $n \in \mathbb{N}$ such that $u_n(x) := z$ for all $x \in I$, for all $n \in \mathbb{N}$ and trivially $u_n \overset{*}{\rightharpoonup} u$ with $u(x) = z$ for all $x \in I$. Using the definition of $H_n$ in (11) and the relation between

$E_n$ and $H_n$ in (10) we obtain

$$\begin{aligned}
E_n(u_n) &= H_n(u_n) - \left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n) \\
&= \frac{1}{2}\sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left| u_n^{i+2} - \frac{\alpha}{2} u_n^{i+1} + u_n^i \right|^2 - \left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n) \\
&= \frac{1}{2}\sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n |z|^2 \left(2 - \frac{\alpha}{2}\right)^2 - \left(1 + \frac{\alpha^2}{8}\right)(1 - c_n\lambda_n) \\
&= (1 - c_n\lambda_n)\left(-1 - \frac{\alpha^2}{8} + \frac{(\alpha-4)^2}{8}\right), \qquad (31)
\end{aligned}$$

for all $n \in \mathbb{N}$. As $(E_n)_n$ $\Gamma-$converges to $E$ by Theorem 4.4, we can make use of the $\liminf$ inequality for $u_n \overset{*}{\rightharpoonup} 0$ as in (29) for

$$\begin{aligned}
f_{\text{hom}}(z) + \left(1 + \frac{\alpha^2}{8}\right) &= \int_I f_{\text{hom}}(z)\,\mathrm{d}x + \left(1 + \frac{\alpha^2}{8}\right) \\
&= E(u) + \left(1 + \frac{\alpha^2}{8}\right) \\
&\leq \liminf_{n\to\infty} E_n(u_n) + \left(1 + \frac{\alpha^2}{8}\right) \\
&= \frac{(\alpha-4)^2}{8}.
\end{aligned}$$

In the last step we used the computation above in (31) for $E_n(u_n)$ and the fact, that $c_n\lambda_n \to 0$ (12).

**Step 2.3.3** We show the upper bound on $B^1$, i.e.,

$$f_{\text{hom}}(z) + \left(1 + \frac{\alpha^2}{8}\right) \leq \frac{(\alpha-4)^2}{8}|z| \text{ for all } z \in B^1.$$

To this end, let $z \in B^1$ and write it as a convex combination of variables on $S^1$, i.e.,

$$z = \beta\hat{z} + (1-\beta)\cdot 0$$

with $\beta := |z| \in [0,1]$ and $\hat{z} := \frac{z}{|z|} \in S^1$. Estimate the upper bound by using the results from Step 2.3.1, Step 2.3.2 and convexity of $f_{\text{hom}}$:

$$\begin{aligned}
f_{\text{hom}}(z) &\leq \beta f_{\text{hom}}(\hat{z}) + (1-\beta) f_{\text{hom}}(0) \\
&\overset{(28)}{=} \beta f_{\text{hom}}(\hat{z}) + (1-\beta)\left(-1 - \frac{\alpha^2}{8}\right) \\
&\overset{(30)}{\leq} \beta\left(\frac{(\alpha-4)^2}{8} - 1 - \frac{\alpha^2}{8}\right) + (1-\beta)\left(-1 - \frac{\alpha^2}{8}\right) \\
&= \beta\frac{(\alpha-4)^2}{8} - 1 - \frac{\alpha^2}{8} \\
&= -\left(1 + \frac{\alpha^2}{8}\right) + \frac{(\alpha-4)^2}{8}|z|,
\end{aligned}$$

which proves the upper bound for $f_{\text{hom}}$. This and the lower bound in Step 2.2 yield (ii).

### Step 2.4

We prove the existence of a convex and non-decreasing function $h : [0,1] \to \mathbb{R}$, such that $f_{\text{hom}}(z) = h(|z|)$ for all $z \in B^1$.

**Step 2.4.1** We show there exists a function $h : [0,1] \to \mathbb{R}$, such that $f_{\text{hom}}(z) = h(|z|)$ for all $z \in B^1$ and $h$ is convex.

To obtain $f_{\text{hom}}$ is radially symmetric, let $R \in SO(2)$ and $z \in B^1$. We define $v^i := Rw^i$ for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor$. and perform a change of variables

$$\begin{aligned}
f_{\text{hom}}(Rz) &= \lim_{\rho\to 0}\lim_{k\to\infty}\frac{1}{k}\inf\left\{-\alpha\sum_{i=0}^{k-2}(v^i, v^{i+1}) + \sum_{i=0}^{k-2}(v^i, v^{i+2}) \mid \langle v\rangle_{1,(0,k)} \in \bar{B}(Rz,\rho)\right\} \\
&= \lim_{\rho\to 0}\lim_{k\to\infty}\frac{1}{k}\inf\left\{-\alpha\sum_{i=0}^{k-2}(Rw^i, Rw^{i+1}) + \sum_{i=0}^{k-2}(Rw^i, Rw^{i+2}) \mid \right. \\
&\qquad\qquad \left. \langle Rw\rangle_{1,(0,k)} \in \bar{B}(Rz,\rho)\right\} \\
&= \lim_{\rho\to 0}\lim_{k\to\infty}\frac{1}{k}\inf\left\{-\alpha\sum_{i=0}^{k-2}(w^i, w^{i+1}) + \sum_{i=0}^{k-2}(w^i, w^{i+2}) \mid \langle w\rangle_{1,(0,k)} \in \bar{B}(z,\rho)\right\} \\
&= f_{\text{hom}}(z).
\end{aligned} \tag{32}$$

This uses that the inner product is invariant under $SO(2)$-transformations and $\langle Rw\rangle_{1,(0,k)} = R\langle w\rangle_{1,(0,k)}$. This justifies writing $f_{\text{hom}}(z) = h(|z|)$ for some $h : [0,1] \to \mathbb{R}$.

Convexity of $h$ follows from the convexity of $f_{\text{hom}}$ and a result in [15, Corollary 12.3.1 and Example below].

**Step 2.4.2** We demonstrate $h$ is non-decreasing.

To show this, let $r_1, r_2 \in [0,1]$ and without loss of generality $r_1 \leq r_2$. Then, choose $z_2 \in B^1$ such that $|z_2| = r_2$ and define the convex combination $z_1 := \frac{r_1}{r_2} z_2 + (1 - r_1/r_2)0$. Since $f_{\text{hom}}$ is convex it follows:

$$f_{\text{hom}}(z_1) \leq \frac{r_1}{r_2} f_{\text{hom}}(z_2) + \left(1 - \frac{r_1}{r_2}\right) f_{\text{hom}}(0) \leq f_{\text{hom}}(z_2).$$

This uses, that $f_{\text{hom}}(0) \leq f_{\text{hom}}(z)$ for all $z \in B^1$. This follows by the lower bound from Step 2.2 (24) and Step 2.3.1 (28) with

$$f_{\text{hom}}(0) \leq \frac{(\alpha-4)^2}{8}|z|^2 + f_{\text{hom}}(0) = \frac{(\alpha-4)^2}{8}|z|^2 - \left(1 + \frac{\alpha^2}{8}\right) \leq f_{\text{hom}}(z)$$

for all $z \in B^1$. By using radial symmetry of $f_{\text{hom}}$, i.e., $f_{\text{hom}}(z) = h(|z|)$ we obtain

$$h(r_1) = h(|z_1|) = f_{\text{hom}}(z_1) \leq f_{\text{hom}}(z_2) = h(r_2).$$

This demonstrates (iii) and finalizes Step 2 of the proof.

## Step 3

We have established the application of the Fundamental Theorem of $\Gamma-$Convergence in Step 1 and characterized the energy density $f_{\text{hom}}$ in Step 2. This brings us to the final part of the proof, where we characterize minimal values and minimizers of $E$.

### Step 3.1

We prove that, for $0 < \alpha \leq 4$,

$$\min_{\mathcal{L}^\infty(I,B^1)} E = E(0) = -\left(1 + \frac{\alpha^2}{8}\right)$$

and $u \equiv 0$ is the unique minimizer in $\mathcal{L}^\infty(I, B^1)$.

To show this, let $\alpha \in (0,4]$.

**Step 3.1.1** We start with $\min_{\mathcal{L}^\infty(I,B^1)} E = E(0) = -\left(1 + \frac{\alpha^2}{8}\right)$.

First, by Step 1 we apply the Fundamental Theorem of $\Gamma-$Convergence 3.12 to $(E_n)_n$. Second, we use the minimal value of $E_n$ for every $n \in \mathbb{N}$ from (13) in

Proposition 4.1 to obtain

$$\begin{aligned}
\min_{\mathcal{L}^\infty(I,B^1)} E &= \lim_{n\to\infty}\left(\inf_{\mathcal{L}^\infty(I,B^1)} E_n\right) \\
&= \lim_{n\to\infty}\left(\min_{\mathcal{L}^\infty(I,B^1)} E_n\right) \\
&= \lim_{n\to\infty}\left(-\left(1+\frac{\alpha^2}{8}\right)(1-c_n\lambda_n)\right) \\
&= -\left(1+\frac{\alpha^2}{8}\right) \\
&= f_{\text{hom}}(0).
\end{aligned} \tag{33}$$

In the last equality we use the value of $f_{\text{hom}}$ at $z=0$ from Step 2.3.1 (28). Then,

$$\min_{\mathcal{L}^\infty(I,B^1)} E = f_{\text{hom}}(0) = \int_I f_{\text{hom}}(0)\,\mathrm{d}x = E(0).$$

**Step 3.1.2** We show $u \equiv 0$ is indeed unique.

To see this, let $z \in B^1\backslash\{0\}$. Then $|z|^2 > 0$. With Step 2.3.1 the value of $f_{\text{hom}}$ is given at 0 as $f_{\text{hom}}(0) = -\left(1+\frac{\alpha^2}{8}\right)$. Taking advantage of the lower bound in Step 2.2 we compute

$$f_{\text{hom}}(z) \geq \frac{(\alpha-4)^2}{8}|z|^2 - \left(1+\frac{\alpha^2}{8}\right) > -\left(1+\frac{\alpha^2}{8}\right) = f_{\text{hom}}(0).$$

Consider $v \in \mathcal{L}^\infty(I,B^1)$ such that $v \neq u$:

$$E(v) = \int_I f_{\text{hom}}(u(x))\,\mathrm{d}x > \int_I f_{\text{hom}}(0)\,\mathrm{d}x = f_{\text{hom}}(0) = E(u).$$

This characterizes the minimal value and minimizer function of $E$ for $0 < \alpha \leq 4$, as claimed in (iv).

**Step 3.2**

We show that, for $\alpha \geq 4$,

$$\begin{aligned}
\min_{\mathcal{L}^\infty(I,B^1)} E = E(u) &= -(\alpha-1) \qquad \text{for all } u \in \mathcal{L}^\infty(I,B^1), \\
f_{\text{hom}}(z) &= -(\alpha-1) \qquad \text{for all } z \in B^1.
\end{aligned}$$

For that purpose, let $\alpha \geq 4$.

**Step 3.2.1** By Step 1 we may apply the Fundamental Theorem of $\Gamma-$Convergence 3.12 and thus we obtain the minimal value of $E$:

$$\min_{\mathcal{L}^\infty(I,B^1)} E = \lim_{n\to\infty}\left(\inf_{\mathcal{L}^\infty(I,B^1)} E_n\right) = \lim_{n\to\infty}\left(-(\alpha-1)(1-c_n\lambda_n)\right) = -\left(\alpha-1\right). \quad (34)$$

**Step 3.2.2** We claim $f_{\text{hom}} = -(\alpha-1)$ on $B^1$.

To show this, first let $z \in S^1$. By Remark 4.2 we know that minimizers $\bar{u}_n$ of $E_n$ are of the form $\bar{u}_n^i = z$ for $i = 0, \dots, \lfloor 1/\lambda_n \rfloor$ and all $n \in \mathbb{N}$. For such a minimizing sequence it holds $\bar{u}_n \overset{*}{\rightharpoonup} \bar{u}$ with $\bar{u}^i := z$ for all $i = 0, \dots, \lfloor 1/\lambda_n \rfloor$. Then, we have

$$\lim_{n\to\infty}\left|E_n(\bar{u}_n) - \inf_{\mathcal{L}^\infty(I,B^1)} E_n\right| = 0.$$

By the Fundamental Theorem of $\Gamma$-convergence 3.12 we have that the limit of $(\bar{u}_n)_n$ is a minimizer of $E$, i.e.,

$$E(\bar{u}) = \min_{\mathcal{L}^\infty(I,B^1)} E \overset{(34)}{=} -(\alpha-1).$$

Then, with

$$E(\bar{u}) = \int_I f_{\text{hom}}(\bar{u}(x))\,dx = \int_I f_{\text{hom}}(z)\,dx = f_{\text{hom}}(z)$$

we have established the value of $f_{\text{hom}}$ on $S^1$.

To show this on $B^1$, let $z \in B^1$ and define $u(x) := z$ for all $x \in I$ such that

$$f_{\text{hom}}(z) = \int_I f_{\text{hom}}(u(x))\,\mathrm{d}x = E(u) \geq \min_{\mathcal{L}^\infty(I,B^1)} E = -(\alpha-1).$$

We conclude with convexity of $f_{\text{hom}}$. For the sake of contradiction assume: $f_{\text{hom}}(z_0) > -(\alpha-1)$ for $z_0 \in B^1 \backslash S^1$. Since $co(S^1) = B^1$ we can write $z_0 = \lambda z_1 + (1-\lambda)z_2$, where $z_1, z_2 \in S^1$ and it holds

$$f_{\text{hom}}(z_0) \leq \lambda f_{\text{hom}}(z_1) + (1-\lambda) f_{\text{hom}}(z_2) = -(\alpha-1),$$

since $f_{\text{hom}}$ is constant on $S^1$. This leads to a contradiction with and therefore

$$f_{\text{hom}}(z) = -(\alpha-1) \text{ for all } z \in B^1. \quad (35)$$

**Step 3.2.3** Finally, we obtain

$$E(u) = \int_I f_{\text{hom}}(u(x))\,\mathrm{d}x \overset{(35)}{=} -(\alpha-1) \qquad \text{for all } u \in \mathcal{L}^\infty(I,B^1).$$

Thus Step 3.2 is done and claim (v) is proven. □

**Remark 4.8** (Lower Semicontinuity of $E$ under the weak* Topology)**.** *The proof of Theorem 4.7 relies on the lower semicontinuity of the $\Gamma$-limit established in Lemma 3.8 to show convexity of $f_{\text{hom}}$. The lemma is stated for metric spaces. The original article [5], however, extends the functionals to $\mathcal{L}^\infty(I,\mathbb{R}^2)$, which is not a metric space with respect to the weak* topology (see also Remark 4.5).*

*We shall nevertheless prove that the functional $\tilde{E} : \mathcal{L}^\infty(I,\mathbb{R}^2) \to (-\infty,+\infty]$ given by*

$$\tilde{E}(u) = \begin{cases} E(u) & \text{if } u \in \mathcal{L}^\infty(I,B^1) \\ +\infty & \text{otherwise in } \mathcal{L}^\infty(I,\mathbb{R}^2) \end{cases}$$

*is lower semicontinuous with respect to the weak* topology on $\mathcal{L}^\infty(I,\mathbb{R}^2)$. The definition of lower semicontinuity (Definition 3.1) can be generalized to topological spaces. For details see [6, Definition 1.1].*

*Proof.* We claim that for all $u \in \mathcal{L}^\infty(I,\mathbb{R}^2)$ and all $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$ it holds

$$E(u) \leq \liminf_{n\to\infty} E(u_n).$$

First, let $u \in \mathcal{L}^\infty(I,B^1)$ be arbitrary and consider a sequence $(u_n)_n \subseteq \mathcal{L}^\infty(I,\mathbb{R}^2)$ such that $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$. There exists a subsequence $(u_{n_k})_k \subseteq (u_n)_n$ such that $\liminf_{n\to\infty} E(u_n) = \lim_{k\to\infty} E(u_{n_k})$. If $\lim_{k\to\infty} E(u_{n_k}) = +\infty$ then

$$\liminf_{n\to\infty} E(u_n) = \lim_{k\to\infty} E(u_{n_k}) = +\infty \geq E(u).$$

If $\lim_{k\to\infty} E(u_{n_k}) < +\infty$ then there exists $(u_{n_{k_l}})_l \subseteq \mathcal{L}^\infty(I,B^1)$ and we obtain

$$\lim_{l\to\infty} E(u_{n_{k_l}}) = \liminf_{n\to\infty} E(u_n) \geq E(u)$$

lower semicontinuity by Lemma 3.8. Second, let $u \in \mathcal{L}^\infty(I,\mathbb{R}^2)\backslash\mathcal{L}^\infty(I,B^1)$, i.e. $\|u\|_\infty > 1$. This implies $E(u) = +\infty$. Let $(u_n)_n \subseteq \mathcal{L}^\infty(I,\mathbb{R}^2)$ be arbitrary such that $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{L}^\infty$, then by the lower semicontinuity of the norm (Proposition 3.2)

$$1 < \|u\|_\infty \leq \liminf_{n\to\infty} \|u_n\|_\infty .$$

This implies that there exists no subsequence in $\mathcal{L}^\infty(I,B^1)$ and hence

$$\liminf_{n\to\infty} E(u_n) = +\infty = E(u).$$

□

## 4.3 Numerical Analysis of the Model

The numerical evaluation and validation were implemented in Python. The complete Python codebase is available on GitHub to ensure reproducibility of the results (see Appendix 6).

For a start, we numerically minimize the energy for a given configuration of a spin chain, reproducing the theoretical results from Proposition 4.1, Theorem 4.4 and Theorem 4.7. Let $u \in \mathcal{C}_n(I, S^1)$ (1) be chosen arbitrarily, i.e.,

$$u^i := \begin{pmatrix} \cos\varphi_i \\ \sin\varphi_i \end{pmatrix}$$

and $\varphi_i \in [0, 2\pi)$ for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor$. Following the computation in (17), we can rewrite the inner products in the energy functional $E_n$ (9) as follows:

$$\begin{aligned} (u_n^i, u_n^{i+1}) &= \cos(\varphi_i - \varphi_{i+1}) \\ (u_n^i, u_n^{i+2}) &= \cos(\varphi_i - \varphi_{i+2}). \end{aligned}$$

Accordingly we obtain for the energy expression:

$$E_n(u) = \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left[ -\alpha \cos(\varphi_i - \varphi_{i+1}) + \cos(\varphi_i - \varphi_{i+2}) \right]$$

and compute the partial derivatives with respect to $\varphi_j$ as

$$\begin{aligned} &\frac{\partial E_n(u)}{\partial \varphi_j} \\ &= \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left[ \alpha \left( \sin(\varphi_i - \varphi_{i+1})(\delta_{ji} - \delta_{j(i+1)} \right) - \left( \sin(\varphi_i - \varphi_{i+2})(\delta_{ij} - \delta_{(i+2)j} \right) \right] \\ &= \lambda_n \left[ \alpha \left( \sin(\varphi_j - \varphi_{j+1}) + \sin(\varphi_j - \varphi_{j-1}) \right) - \left( \sin(\varphi_j - \varphi_{j+2}) + \sin(\varphi_j - \varphi_{j-2}) \right) \right], \end{aligned}$$

for $j = 0, \ldots, \lfloor 1/\lambda_n \rfloor - 2$. For spin indices at or near the boundary of the lattice, any term in this expression that refers to an index falling outside the defined range $\mathbb{Z}_n$ is omitted.

For the numerical analysis we choose $\lambda_n := 1/n$, $n \in \mathbb{N}$. The method used for minimization is Sequential Least Squares Programming (SLSQP), which is a variant of Sequential Quadratic Programming (SQP) and is provided by [19, SciPy]. This method allows for a target function $E_n$, gradient function $\nabla E_n$ and a constraint, here the periodic boundary condition from (8).

### Helimagnet

Consider $0 < \alpha \leq 4$. In a first step the energy $\min E_n$ is minimized numerically for $n = 1, ..., 200$. For each $n$ we get a minimizing state $u_n^{\min}$, such that $E_n(u_n^{\min}) \approx \min E_n$. Three different inital states are passed to the optimizer: a random spin chain (Fig. 4), the optimal spin chain with a small perturbation (Fig. 5) and the optimal spin chain without any perturbation (Fig. 6).

The optimal minimizing spin chain is as characterized in Proposition 4.1 (14) by an angle satisfying $\varphi^* = \arccos(\alpha/4)$. For the slightly perturbed spin chain the angles are $\tilde{\varphi} := \varphi^* + \delta$ and $\delta$ is drawn from a uniform distribution on the interval $[0, \varepsilon)$. As reference the minimal energy $\min E_n = -(1 + \alpha^2/8)(1 - c_n\lambda_n)$ from Proposition 4.1 (13) and the minimum value of the $\Gamma-$limit energy $\min E = -(1 + \alpha^2/8)$ from Theorem 4.7 (iv) are plotted as well.

One observes that random initial spin chains frequently fail to reach the theoretical minimum, indicating existence of local minima. This is demonstrated in Fig. 4. In contrast, Fig. 5 shows that starting from a slightly perturbed ground state avoids these local minima and this approach consistently converges to the theoretical ground state energy. The microscopic origin of this excess energy for random initial configurations is revealed in Fig. 9. The direction of the spin rotation flips multiple times. These defects are energetically costly preventing the system from reaching the true ground state without any flips, which is shown in that figure as well.

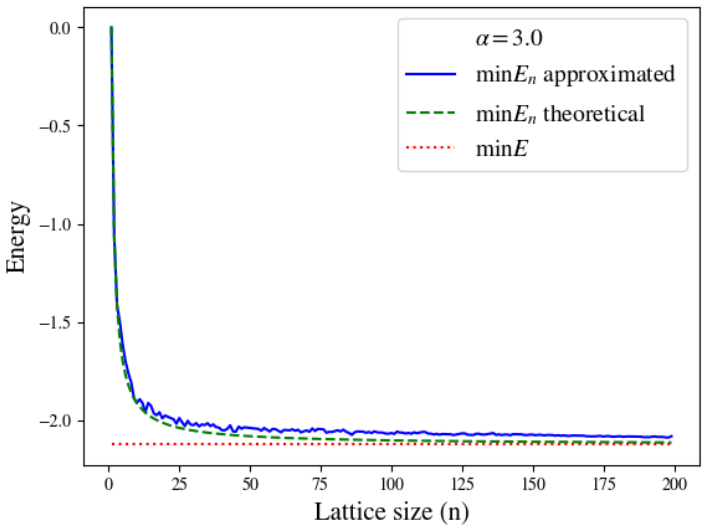


Figure 4: Convergence of the numerically minimized energy for lattice sizes $n = 1, \ldots, 200$. The minimization process is initialized with a random spin configuration for each $n$. Interaction ratio $\alpha = 3$. One observes the minimization getting trapped in local minima.

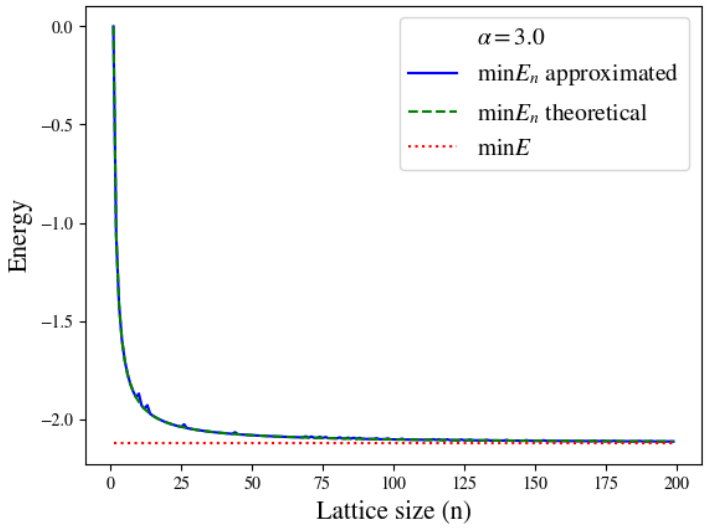


Figure 5: Convergence of the numerically minimized energy for lattice sizes $n = 1, \ldots, 200$. The minimization process is initialized with a slightly perturbed optimal spin chain (perturbation of order $\varepsilon = 3$). Interaction ratio $\alpha = 3$. The minimization consistently converges to the theoretical minimum.

This reproduces two things: First, the value of $\min E_n$ in (13) from Proposition 4.1. And second, the application of the Fundamental Theorem of $\Gamma-$Convergence 3.12 in the proof of Theorem 4.7 (iv), where $\min E = \lim_{n\to\infty}(\min E_n)$ (33).

Let us have a closer look at minimizing states. Consider $u_{50}^{\min}$ from the minimization of $E_{50}$. The spin chains in Fig. 7 and Fig. 8 correspond to the random initialization and the perturbed optimal initialization of the energy convergence process above. The figures show the spin chain that was given initially to the optimizer, the spin chain that was returned from the optimizer and as a reference the theoretical optimal spin chain from Proposition 4.1. This reproduces the helimagnetic pattern of the optimal minimizing spin chain as proven in Proposition 4.1 (14), and visualizes a spin chain in a local minimum.

One can analyze the chirality of the helimagnetic states, i.e., determine whether the spins rotate clockwise (negative) or counter-clockwise (positive). We represent this symmetry property by mapping the angle between nearest-neighbors to $\{-1, 1\}$ via, $\operatorname{sign}(\sin(\varphi_{i+1} - \varphi_i))$ for $i = 0, \ldots \lfloor 1/\lambda_n \rfloor - 1$, that is, positive or negative chirality. The introduction to chirality is to give the reader an intuition for the next section, where the helimagnet/ferromagnet transition point is analyzed, and chirality plays a crucial role.

Then, Fig. 9 analyzes $u_{50}^{\min}$ in the three different cases of initialization. For the theoretical minimizer no chirality flips appear. For completness of the discussion it should be said, that, up to global rotation (cf. discussion of rotational invariance

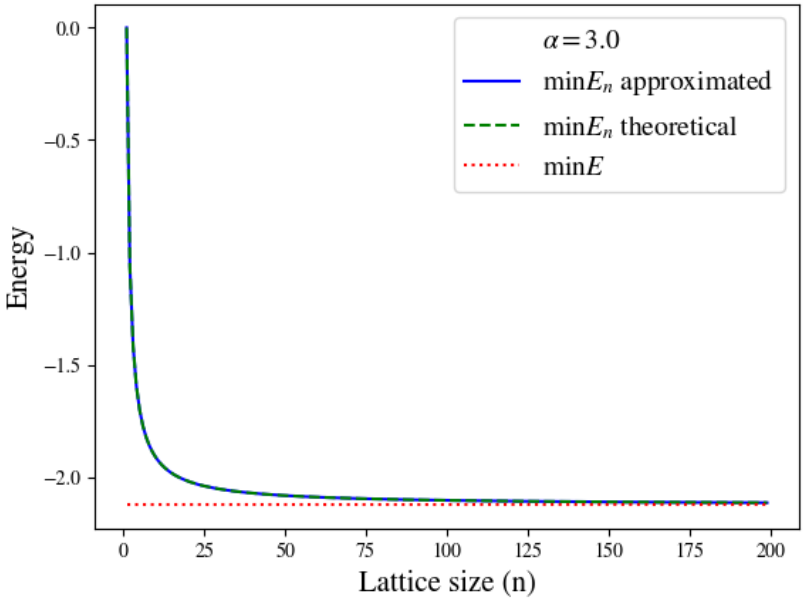


Figure 6: Convergence of the numerically minimized energy for lattice sizes $n = 1, \ldots, 200$. The minimization process is initialized with the exact theoretical optimal minimizing spin chain. Interaction ratio $\alpha = 3$. The minimization is observed to be in a global minimum, as expected.

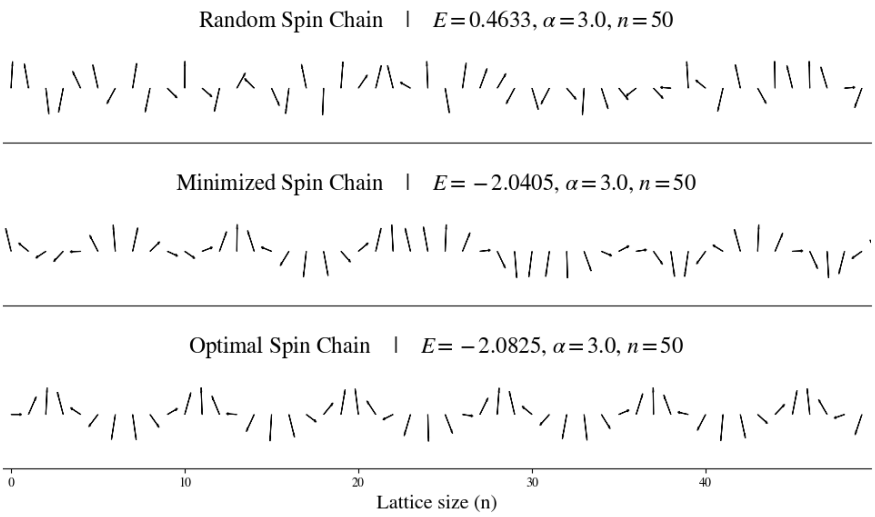


Figure 7: Comparison of spin configurations for lattice size $n = 50$ in the helimagnetic regime $\alpha = 3$. (Top) The initial random spin chain. (Middle) The numerically minimizing spin chain. (Bottom) The true minimizing spin chain. For reference: the energy value for each spin chain. Notice the difference in energy and imperfection of the helical pattern.

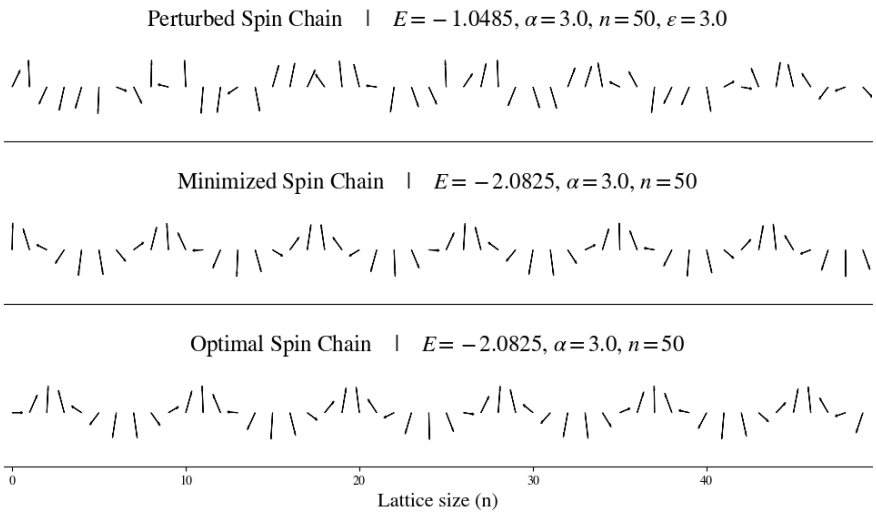


Figure 8: Comparison of spin configurations for lattice size $n = 50$ in the helimagnetic regime $\alpha = 3$. (Top) The initial perturbed spin chain. (Middle) The numerically minimizing spin chain. (Bottom) The true minimizing spin chain. For reference: the energy value for each spin chain. Notice consistency with the optimal spin chain.

of $f_{\text{hom}}$ in (32)), there exist two minimizing configurations. The one with positive chirality is displayed in Fig. 9.

### Ferromagnet

Consider $\alpha \geq 4$. According to Remark 4.2 the minimizing states are ferromagnetic, i.e., aligned. The minimization process runs smoothly, therefore only $u_{50}^{\min}$ with random initial spin chain is depicted in Fig. 10.

This reproduces the value of $\min E_n$ in Remark 4.2 and the ferromagnetic pattern of the optimal minimizing spin chain in Remark 4.2.

Thus far, the analysis concentrated on the minimal energy. However, Theorem 4.4 provides the limit energy for an arbitrary state variable $u$. To evaluate the intregral, $f_{\text{hom}}$ is needed, for which we are given a homogenization formula, as well as an upper and lower bound from Theorem 4.7. In Fig. 11 a numerical approximation of $f_{\text{hom}}$ is displayed. The limit in the homogenization formula was set $k = 300$, nonetheless it should be noted that the approximation breaches the upper bound. This indicates that stand-alone numerical solutions should be interpreted with caution and ideally supported by analytical verification. This finalizes the numerical illustration of the theoretical results of Theorem 4.4 and Theorem 4.7.

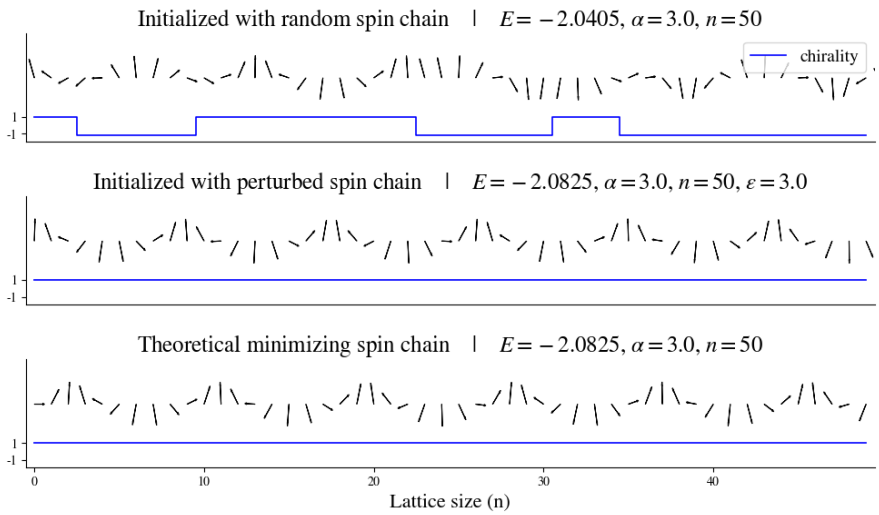


Figure 9: Chirality analysis of the minimized states for lattice size $n = 50$ in the helimagnetic regime $\alpha = 3$. (Top) Minimizing configuration resulting from random initial configuration. (Middle) Minimizing configuration resulting from perturbed intial configuration. (Bottom) Theoretical minimizing spin configuration. For reference: the chirality for each spin chain. Notice the chirality flips for random initial spin chains.

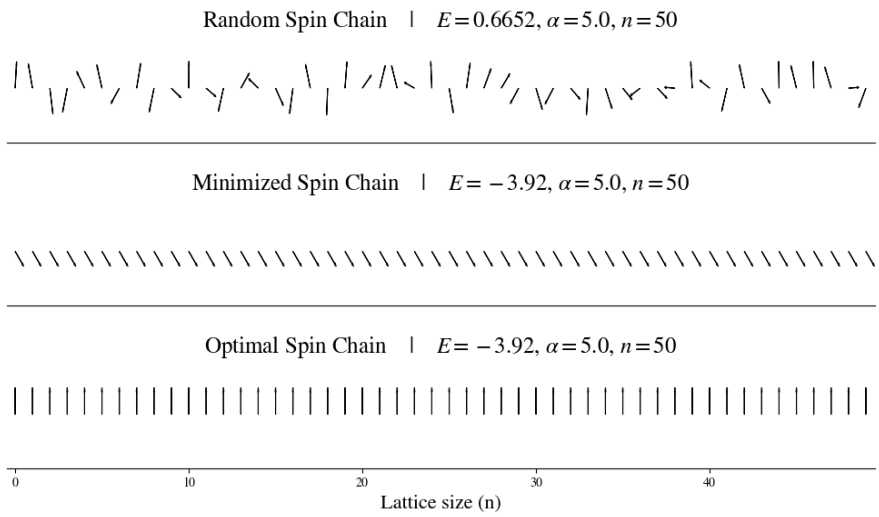


Figure 10: Comparison of spin configurations for lattice size $n = 50$ in the ferromagnetic regime $\alpha = 5$. (Top) Initial random configuration. (Middle) Minimizing configuration resulting from minimization process. (Bottom) Theoretical minimizing spin configuration. For reference: the energy value for each spin chain. The global minimum is attaind perfectly.

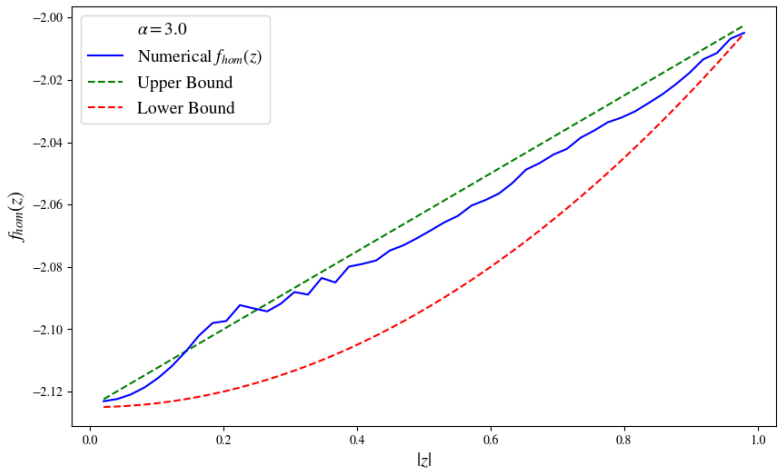


Figure 11: The numerical approximation of the homogenization formula for $f_{\text{hom}}$ from Theorem 4.4 enclosed by its upper and lower bounds. The numerical approximation slightly exceeds the theoretical upper bound due to discretization effects. The limit parameters used here are $k = 300$ and $\rho = 0$.

## 4.4 Chirality Transitions

This section treats the helimagnetic/ferromagnetic transition point. Precise defintions and detailed proofs can be found in [5, Chapter 4]. The transition point ($\alpha \simeq 4$) has been established in Proposition 4.1 and Remark 4.2. Let the interaction ratio $\alpha$ be scale dependent, approching 4 from below, i.e., $\alpha_n = 4(1 - \delta_n)$ with $\delta_n$ a vanishing sequence of positive numbers, so $\alpha_n$ is close to the transition point. We introduce a new functional whose asymptotic behavior will better describe the ground states of the system. Define $E_n^{hf} : \mathcal{L}^\infty(I, B^1) \to (-\infty, +\infty]$ and $H_n^{hf} : \mathcal{L}^\infty(I, B^1) \to [0, +\infty]$ by plugging the scaled interaction ratio $\alpha_n$ into the energies $E_n$ in (9) and $H_n$ in (11):

$$E_n^{hf}(u) := \begin{cases} -4(1-\delta_n) \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+1}) + \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n(u^i, u^{i+2}) & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise,} \end{cases}$$

$$H_n^{hf}(u) := \begin{cases} \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n |u^{i+2} - 2(1-\delta_n)u^{i+1} + u^i|^2 & \text{if } u \in \mathcal{C}_n(I, S^1) \\ +\infty & \text{otherwise.} \end{cases}$$

The properties proven so far in Section 4 remain valid for these new functionals. In particular by Theorem 4.7 and (10) it holds

$$H_n^{hf}(u) = E_n^{hf}(u) - \min E_n^{hf} \tag{36}$$

Going from the functional $H_n^{hf}$, we will perform a formal computation leading to the $\Gamma-$limit [5, Introduction]. On that point consider $u \in \mathcal{C}_n(I, S^1)$. For $\lambda_n$ small enough we may use the central discrete difference approximating the second derivative $(u^{i+2} - 2u^{i+1} + u^i)/\lambda_n^2 \simeq u''(\lambda_n(i+1))$ and expand the square as follows

$$\begin{aligned}
H_n^{hf}(u) &= \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left| u^{i+2} - 2(1-\delta_n)u^{i+1} + u^i \right|^2 \\
&= \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left| \lambda_n^2 \left( \frac{u^{i+2} - 2u^{i+1} + u^i}{\lambda_n^2} \right) + 2\delta_n u^{i+1} \right|^2 \\
&\simeq \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left| \lambda_n^2 u''(\lambda_n(i+1)) + 2\delta_n u^{i+1} \right|^2 \\
&\simeq \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 2} \lambda_n \left( \lambda_n^4 |u''(\lambda_n(i+1))|^2 + 4\delta_n^2 \underbrace{|u^{i+1}|^2}_{=1} + 4\lambda_n^2 \delta_n u''(\lambda_n(i+1)) u^{i+1} \right) \\
&\simeq \frac{1}{2} \int_I \left( \lambda_n^4 |u''(x)|^2 + 4\delta_n^2 + 4\lambda_n^2 \delta_n u''(x) u(x) \right) \mathrm{d}x.
\end{aligned}$$

In the last step we have substituted the sum by an integral, again assuming $\lambda_n$ to be small. We integrate by parts

$$H_n^{hf}(u) \simeq \frac{1}{2} \int_I \left( \lambda_n^4 |u''(x)|^2 + 4\delta_n^2 \right) \mathrm{d}x + \frac{1}{2} 4\lambda_n^2 \delta_n \left( \underbrace{u'(x)u(x)\Big|_0^1}_{=0} - \int_I |u'(x)|^2 \, \mathrm{d}x \right)$$

and claim the boundary term vanishes. In the discrete case we compute the backwards difference of the first derivative

$$u'(x)u(x) \simeq u^{i+1}(u^{i+1} - u^i)/\lambda_n \simeq (1 - (u^{i+1}, u^i))/\lambda_n$$

and the periodic boundary condition $(u^{\lfloor 1/\lambda_n \rfloor}, u^{\lfloor 1/\lambda_n \rfloor - 1}) = (u^1, u^0)$ confirms the claim. We assume that there exists $\varphi : \mathbb{R} \to [0, 2\pi)$ with sufficient regularity such that we can write

$$u(x) = \begin{pmatrix} \cos\varphi \\ \sin\varphi \end{pmatrix} \in S^1$$

and introduce the angular velocity $\theta(x) := \varphi'(x)$ in $I$. Indeed $u$ and $u'$ are orthogonal and the boundary term vanishes in the continuous setting as expected. The functional can be rewritten in terms of $\theta$, thereto we first compute the norm of the derivates

$$|u'|^2 = |\varphi' \begin{pmatrix} -\sin\varphi \\ \cos\varphi \end{pmatrix}|^2 = \theta^2,$$
$$|u''(x)|^2 = |\varphi'' \begin{pmatrix} -\sin\varphi \\ \cos\varphi \end{pmatrix} + (\varphi')^2 \begin{pmatrix} -\cos\varphi \\ -\sin\varphi \end{pmatrix}|^2 = (\theta')^2 + \theta^4 + \varphi''\varphi \cdot 0$$

and

$$\begin{aligned} H_n^{hf}(\theta) &= \frac{1}{2} \int_I \left(\lambda_n^4(\theta'(x)^2 + \lambda_n^4\theta^4 + 4\delta_n^2 - 4\lambda_n^2\delta_n\theta^2\right) \mathrm{d}x \\ &= \frac{1}{2} \int_I \left(\lambda_n^4(\theta'(x))^2 + (\lambda_n^2\theta(x)^2 - 2\delta_n)^2\right) \mathrm{d}x. \end{aligned}$$

We define the scaled angular velocity as $z := \varepsilon_n\theta$ with first derivatice $z' = \varepsilon_n\theta'$ where $\varepsilon_n := \frac{\lambda_n}{\sqrt{2\delta_n}}$ and regard the functional as defined on $z \in \mathcal{L}^1(I, \mathbb{R})$

$$H_n^{hf}(z) = \int_I \left(\lambda_n^2\delta_n(z'(x))^2 + 2\delta_n^2(z(x)^2 - 1)^2\right) \mathrm{d}x.$$

We rescale the functional with $\sqrt{2}\lambda_n\delta_n^{3/2}$ leaving us with what is called a Modica-Mortola-type functional in the literature:

$$\frac{H_n^{hf}(z)}{\sqrt{2}\lambda_n\delta_n^{3/2}} = \int_I \left(\varepsilon_n(z'(x))^2 + \frac{1}{\varepsilon_n}(z(x)^2 - 1)^2\right) \mathrm{d}x, \tag{37}$$

Considering (36) and (37), it becomes clear that we have zoomed in such that

$$\frac{H_n^{hf}(z)}{\sqrt{2}\lambda_n\delta_n^{3/2}} \simeq \frac{E_n^{hf}(u) - \min E_n^{hf}}{\sqrt{2}\lambda_n\delta_n^{3/2}}.$$

On this scale the chirality transition can be observed. The frustration has translated into competing terms of $z$ and $z'$. The $z$-term favors chirality flips, i.e., functions jumping from $+1$ to $-1$. In contrary the $z'$-term penalizes these flips and favors a constant $z$. We think it is advisable to examine a result that can be proven, describing the $\Gamma-$limit $H^{hf}(z) := \Gamma - \lim_{n\to\infty} H_n^{hf}(z)/\sqrt{2}\lambda_n\delta_n^{3/2}$, depending on the asymptotic behavior of the scaling parameters $l := \lim_{n\to\infty} \varepsilon_n = \lim_{n\to\infty} \lambda_n/\sqrt{2\delta_n}$. As we will see, the behavior of the rigorous $\Gamma$-limit is determined by the competing terms in expression (37).

To be able to present the exact statement, we need more precise notation for the

order parameter $z$. Given $u_n \in \mathcal{C}_n(I, S^1)$ we define for $i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 1$ the oriented central angle $\theta_n^i \in [-\pi, \pi)$ as

$$\theta_n^i := \chi[u_n^i, u_n^{i+1}] \arccos((u_n^i, u_n^{i+1})),$$

where $\chi : S^1 \times S^1 \to \{\pm 1\}$, $\chi[v, w] := \operatorname{sign}(v_1 w_2 - v_2 w_1)$ with the convenction that $\operatorname{sign}(0) = -1$. Then, we define the set of order parameters as

$$\begin{aligned}\tilde{\mathcal{C}}_n(I, \mathbb{R}) := \{z : [0,1) \to \mathbb{R} \mid z(x) = \sqrt{2/\delta_n} \sin(\theta_n^i/2), \text{ if } x \in \lambda_n(i + [0,1)),\\ i = 0, \dots, \lfloor 1/\lambda_n \rfloor - 1\}.\end{aligned} \tag{38}$$

With the above definitions the following functional $H_n^{hf} : \mathcal{L}^1(I, \mathbb{R}) \to [0, +\infty]$ penalizes the helimagnetic transition around the ferromagnetic state.

$$H_n^{hf}(z) := \begin{cases} H_n^{hf}(u) & \text{if } z \in \tilde{\mathcal{C}}_n(I, \mathbb{R}) \\ +\infty & \text{otherwise} \end{cases} \tag{39}$$

The rigorous justifications of these definitons can be found in [5, Chapter 4].

**Theorem 4.9** ($\Gamma$-convergence of the Chiral Energy)**.** *[5, Theorem 4.2]*
*Let $H_n^{hf} : \mathcal{L}^1(I, \mathbb{R}) \to [0, +\infty]$ be given by*

$$H_n^{hf}(z) = \begin{cases} H_n^{hf}(u) & \textit{if } z \in \tilde{\mathcal{C}}_n(I, \mathbb{R}) \\ +\infty & \textit{otherwise} \end{cases}$$

*as in (39) and $\tilde{\mathcal{C}}_n(I, \mathbb{R})$ as in (38). Assume that the limit $l := \lim_{n\to\infty} \lambda_n/\sqrt{2\delta_n}$ exists. Then $H^{hf}(z) := \Gamma - \lim_{n\to\infty} H_n^{hf}(z)/\sqrt{2}\lambda_n \delta_n^{3/2}$ with respect to $\mathcal{L}^1(I)$ convergence is given by one of the following formulas:*

*(i) if $l = 0$*

$$H^{hf}(z) := \begin{cases} \frac{8}{3}\#(S(z)) & \textit{if } z \in BV(I, \{\pm 1\}), \\ +\infty & \textit{otherwise}, \end{cases}$$

*where $S(z)$ is the set of jumps of $z$.*

*(ii) if $l \in (0, +\infty)$*

$$H^{hf}(z) := \begin{cases} l \int_I (z'(x))^2 \,\mathrm{d}x + \frac{1}{l} \int_I (z^2(x) - 1)^2 \,\mathrm{d}x & \textit{if } z \in W^{1,2}_{|per|}(I), \\ +\infty & \textit{otherwise}, \end{cases}$$

*where we have set $W^{1,2}_{|per|}(I) := \{z \in W^{1,2} \mid |z(0)| = |z(1)|\}$.*

*(iii) if* $l = +\infty$

$$H^{hf}(z) := \begin{cases} 0 & \textit{if } z = \textit{const}, \\ +\infty & \textit{otherwise}. \end{cases}$$

*Proof.* Can be found in the given reference. □

Notice that $H^{hf}$ in the case (ii) $l \in (0, +\infty)$ corresponds to the result of our heuristic in (37). This expression helps analyzing the behavior in the three cases.

When $l = \lim_{n\to\infty} \lambda_n/\sqrt{2\delta_n} = 0$, this corresponds to the scaling where the lattice spacing $\lambda_n$ is much smaller than the frustration parameter $\delta_n$. In the integral expression in (ii) the second term ($z$-term) is the one potentially contributing to the functional. The only case where the term does not diverge is if $z(x) \in \{\pm 1\}$. In this case the system may have chirality transitions, represented through the order parameter z. If the variation of $z$ is bounded the set $S(z)$ is finite and counts the number of jumps of the chirality of the system. The order parameter $z$ is then a $BV-$function, see Example 2.3.

In the case $l \in (0, +\infty)$ there is a balance between the ferromagnetic NN interaction and the antiferromagnetic NNN interaction, leading to modulated ground states. The system may have chirality transitions with finite limit energy on $W^{1,2}_{|per|}$, this corresponds to the assumptions we have made in the heuristic computation and which are needed for the integral to be finite.

The last case $l = \lim_{n\to\infty} \lambda_n/\sqrt{2\delta_n} = +\infty$ corresponds to the scaling where the frustration parameter $\delta_n$ is much smaller than the lattice size $\lambda_n$. The system effectively becomes ferromagnetic. Again looking at the integral expression in (ii) we find the first term ($z'$-term) potentially contributing to the functional. The only case were the term does not diverge is if $z'(x) \equiv 0$, which implies that z is indeed constant. Hence chirality transitions are energetically forbidden.

# 5 Spin Chain Model of Higher-Order Interactions

The F-AF model is taking into consideration the nearest neighbor (NN) and next-to-nearest neighbor (NN) interaction. Real materials often exhibit higher-order interactions beyond NN and NNN (e.g., in cuprates), and including NNNN captures these physical effects more realistically (see [8, Chapter 4]). In this Chapter we expand the model from Section 4 by additionally taking into account the contribution of the next-to-next-to-nearest neighbor (NNNN) interaction with a ferromagnetic $(-)$ or antiferromagnetic $(+)$ interaction parameter $J_3 > 0$. We call this system F-AF-F, respectively F-AF-AF spin chain model. We keep the notation introduced for the F-AF spin chain model in Section 4. Then, the energy for the extended model is

$E_n^{\pm} : \mathcal{U}_n(I) \to \mathbb{R}$

$$E_n^{\pm}(u) = \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n \left( -J_1(u^i, u^{i+1}) + J_2(u^i, u^{i+2}) \pm J_3(u^i, u^{i+3}) \right).$$

We simplify in dividing the energy by $J_3$ and set $E_n^{\pm} = E_n^{\pm}/J_3$. Introducing interaction ratios $\alpha_1 := J_1/J_3$ and $\alpha_2 := J_2/J_3$ yields

$$E_n^{\pm}(u) = \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n \left( -\alpha_1(u^i, u^{i+1}) + \alpha_2(u^i, u^{i+2}) \pm (u^i, u^{i+3}) \right). \tag{40}$$

Furthermore, we fix periodic boundary conditions on the system, to avoid edge artifacts:

$$\begin{aligned} (u^0, u^1) &= (u^{\lfloor 1/\lambda_n \rfloor - 2}, u^{\lfloor 1/\lambda_n \rfloor - 1}) \\ (u^1, u^2) &= (u^{\lfloor 1/\lambda_n \rfloor - 1}, u^{\lfloor 1/\lambda_n \rfloor}) \\ (u^0, u^2) &= (u^{\lfloor 1/\lambda_n \rfloor - 2}, u^{\lfloor 1/\lambda_n \rfloor}) \end{aligned} \tag{41}$$

and write for these functions for simplicity of notation $u \in \bar{\mathcal{U}}_n(I, S^1)$ (1). Following the F-AF model in (10) we reformulate $E_n^{\pm}$ with help of a functional $H_n^{\pm}$. The subsequent analysis of its properties are more tractable. We define $H_n^{\pm} : \bar{\mathcal{U}}_n(I, S^1) \to [0, \infty)$ as

$$H_n^{\pm}(u) := \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n \left| u^{i+3} + \gamma^{\pm} u^{i+2} + \beta^{\pm} u^{i+1} \pm u^i \right|^2 \tag{42}$$

where $\gamma^{\pm}$ and $\beta^{\pm}$ are given by

$$\begin{aligned} \gamma^{\pm} &:= \pm \frac{\alpha_2}{2} + \sqrt{\frac{\alpha_2^2}{4} + \alpha_2 \pm \alpha_1} \\ \beta^{\pm} &:= \frac{\alpha_2}{2} \mp \sqrt{\frac{\alpha_2^2}{4} + \alpha_2 \pm \alpha_1} \end{aligned} \tag{43}$$

For $u \in \bar{\mathcal{U}}_n(I, S^1)$ disolve the quadratic summand and notice that $|u^i|^2 = 1$ for all $i = 0, \ldots, \lfloor 1/\lambda_n \rfloor$

$$\begin{aligned} &\left| u^{i+3} + \gamma^{\pm} u^{i+2} + \beta^{\pm} u^{i+1} \pm u^i \right|^2 \\ &= 2 + (\gamma^{\pm})^2 + (\beta^{\pm})^2 \\ &+ 2\gamma^{\pm}(u^{i+2}, u^{i+3}) + 2\gamma^{\pm}\beta^{\pm}(u^{i+1}, u^{i+2}) \pm 2\beta^{\pm}(u^{i+1}, u^i) \\ &+ 2\beta^{\pm}(u^{i+3}, u^{i+1}) \pm 2\gamma^{\pm}(u^{i+2}, u^i) \pm 2(u^{i+3}, u^i). \end{aligned}$$

With the periodic boundary conditions (41) we can perform index shifts and unite the NN, NNN and NNNN terms such that:

$$H_n(u) = \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n (2 + (\gamma^\pm)^2 + (\beta^\pm)^2 + (u^{i+1}, u^i)(2\gamma^\pm + 2\gamma^\pm\beta^\pm \pm 2\beta^\pm) \\ + (u^{i+2}, u^i)(2\beta^\pm \pm 2\gamma^\pm) \pm 2(u^{i+3}, u^i))$$

and we compute the prefactors

$$\begin{aligned} \frac{1}{2}(2 + (\gamma^\pm)^2 + (\beta^\pm)^2) &= 1 + \frac{\alpha_2^2}{2} + \alpha_2 \pm \alpha_1 \\ \frac{1}{2}(2\gamma^\pm + 2\gamma^\pm\beta^\pm \pm 2\beta^\pm) &= -\alpha_1 \\ \frac{1}{2}(2\beta^\pm \pm 2\gamma^\pm) &= \alpha_2. \end{aligned}$$

To simplify the representation define $\omega^\pm := 1 + \frac{\alpha_2^2}{2} + \alpha_2 \pm \alpha_1$ and $d_n := \frac{1}{\lambda_n} - \left\lfloor \frac{1}{\lambda_n} \right\rfloor + 2 \in [2, 3)$, so that

$$\sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n = 1 - 1 + \lambda_n(\lfloor 1/\lambda_n \rfloor - 2) = 1 - d_n\lambda_n \xrightarrow{n\to\infty} 1.$$

Then we can write $H_n^\pm$ in terms of $E_n^\pm$ as

$$H_n^\pm(u) = E_n^\pm(u) + (1 - d_n\lambda_n)\omega^\pm \qquad \text{for all } u \in \bar{\mathcal{U}}_n(I, S^1). \tag{44}$$

Having established this relation between $E_n^\pm$ and $H_n^\pm$, we continue by analyzing the properties of $H_n^\pm$. In a first result we prove a lower bound for $H_n^\pm$ and subsequently obtain as a Corollary bounds on the infimum of the energy $E_n^\pm$.

**Proposition 5.1** (Lower Bound for the Auxiliary Energy)**.** *Let $\alpha_1 > 0$, $\alpha_2 > 0$ and $\gamma^\pm, \beta^\pm$ as in (43). Let $H_n^\pm : \bar{\mathcal{U}}_n(I, S^1) \to [0, \infty)$ for $n \in \mathbb{N}$ be given by*

$$H_n^\pm(u) := \frac{1}{2} \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n \left| u^{i+3} + \gamma^\pm u^{i+2} + \beta^\pm u^{i+1} \pm u^i \right|^2$$

*as in (42). Further we define $P^\pm : \mathbb{C} \to \mathbb{C}$ as*

$$P^\pm(z) := z^3 + \gamma^\pm z^2 + \beta^\pm z \pm 1,$$

*and set*

$$P_{n,\min}^\pm := \min_{j=0,\dots,\lfloor 1/\lambda_n \rfloor - 3} \frac{1}{2} \left| P^\pm \left( \exp\left( \frac{2\pi i}{\lfloor 1/\lambda_n \rfloor - 2} j \right) \right) \right|^2.$$

*Then for all $u \in \bar{\mathcal{U}}_n(I, S^1)$ we have*

$$H_n^{\pm}(u) \geq P_{n,\min}^{\pm}(1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + |\gamma^{\pm}| + |\beta^{\pm}|)^2.$$

*Proof.* Let $\alpha_1 > 0$ and $\alpha_2 > 0$. Set $N := \lfloor 1/\lambda_n \rfloor - 2$. Let $H_n^{\pm}$ be defined as in (42). For legibility reasons we abandon the F/AF distinction ($\pm$) and prove the Proposition for $H_n^+$ only. The proof for $H_n^-$ is analogous and in the one part where the sign matters, it will be mentioned.

We have to show that for all $u \in \bar{\mathcal{U}}_n(I, S^1)$

$$H_n(u) \geq \min_{j=0,\dots,N-1} \frac{1}{2}\left|P\left(\exp\left(\frac{2\pi i}{N} j\right)\right)\right|^2 (1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + \left|\gamma^{\pm}\right| + \left|\beta^{\pm}\right|)^2.$$

First, we consider the problem in a complex setting. Then we split $H_n$ into a bulk part and a boundary part. We find estimates for each part and finally obtain a lower bound for $H_n$.

**Step 1** We transfer the problem to a new setting. Identifiy $S^1$ with the unit circle in $\mathbb{C}$ via

$$\begin{pmatrix} \cos\varphi \\ \sin\varphi \end{pmatrix} \in S^1 \mapsto \exp(i\varphi) \in \mathbb{C},$$

with $|\exp(i\varphi)|^2 = 1$. We will now view the function $u$ as a function $u \in \bar{\mathcal{U}}_n(I, \mathbb{C})$. For convenience purposes define $b \in \mathcal{U}_n(I, \mathbb{C})$ for $u \in \bar{\mathcal{U}}_n(I, \mathbb{C})$ as

$$b(t) := u^{t+3} + \gamma u^{t+2} + \beta u^{t+1} + u^t$$

for $t = 0, \dots, N-1$, so that we can write $H_n(u) = \frac{1}{2} \sum_{t=0}^{N-1} \lambda_n |b(t)|^2$.

Now, we split $H_n$ into a bulk part with indices $\{0, \dots, N-4\}$ and a boundary part with indices $\{N-3, N-2, N-1\}$ and find estimates for each of them. To explain the choice of bulk and boundary: The index $t_* := N - 4$ corresponds to the range of the sum in $H_n$, since the highest index in $b(t)$ is $t + 3$ and thus $t_* + 3 = N - 1$. Higher indices are considered as boundary.

**Step 2** We estimate the bulk part $H_n^{\text{per}}$ from below:

$$H_n^{\text{per}}(v) \geq P_{n,\min}(1 - d_n\lambda_n).$$

To show this and to define $H_n^{\text{per}}$, we employ Properties of the Discrete Fourier Transform (Proposition 2.13) and consider the $N-$periodic sequence space $\mathbb{C}^N$ therein. We define a subspace

$$\mathbb{C}_1^N := \{f \in \mathbb{C}^N \mid |f(j)| = 1 \text{ for all } j = 0, \dots, N-1\}.$$

For $u \in \bar{\mathcal{U}}_n(I, \mathbb{C})$ define the truncated $u$ as a $N-$periodic sequence $v(t) \coloneqq u^t$ for $t = 0, \ldots, N-1$ such that $v \in \mathbb{C}_1^N$. Further we define the $N-$periodic sequence $c \in \mathbb{C}_1^N$ as

$$c(t) \coloneqq v(t + 3 \bmod N) + \gamma v(t + 2 \bmod N) + \beta v(t + 1 \bmod N) + v(t \bmod N).$$

The corresponding expression for the energy $H_n^{\mathrm{per}} : \mathbb{C}_1^N \to [0, +\infty]$ is denoted by

$$H_n^{\mathrm{per}}(v) \coloneqq \frac{1}{2} \sum_{t=0}^{N-1} \lambda_n |c(t)|^2.$$

The following sequence shall be defined distincting the F/AF case, to make the proof clear. We define $k^{\pm} \in \mathbb{C}_1^N$ by

$$k^{\pm}(t) \coloneqq \begin{cases} 1 & \text{if } t = N-3 \\ \gamma & \text{if } t = N-2 \\ \beta & \text{if } t = N-1 \\ \pm 1 & \text{if } t = 0 \\ 0 & \text{otherwise} \end{cases}$$

and continue the proof with $k \coloneqq k^+$. We can write c as a convolution ($\star$) of $k$ and $v$ following (iii) in Proposition 2.13:

$$c(t) = \sum_{s=0}^{N-1} k(s) v(t-s) = N(k \star v)(t).$$

Then, we obtain a first lower bound for the bulk part using Plancherel Identity (P.I.)

from (iv) Proposition 2.13:

$$
\begin{aligned}
H_n^{\mathrm{per}}(v) &= \frac{1}{2}\sum_{t=0}^{N-1}\lambda_n|c(t)|^2 \\
&\overset{(P.I.)}{=} \frac{1}{2}N\sum_{t=0}^{N-1}\lambda_n|\hat{c}(t)|^2 \\
&= \frac{1}{2}N\sum_{t=0}^{N-1}\lambda_n|N\widehat{(k\star v)}(t)|^2 \\
&= \frac{1}{2}N\sum_{t=0}^{N-1}\lambda_n|N\hat{k}(t)\hat{v}(t)|^2 \\
&\geq \min_{t=0,\dots,N-1}|N\hat{k}(t)|^2\frac{1}{2}N\sum_{t=0}^{N-1}\lambda_n|\hat{v}(t)|^2 \\
&\overset{(P.I.)}{=} \min_{t=0,\dots,N-1}|N\hat{k}(t)|^2\frac{1}{2}\sum_{t=0}^{N-1}\lambda_n\underbrace{|v(t)|^2}_{=1} \\
&= \min_{t=0,\dots,N-1}\frac{1}{2}|N\hat{k}(t)|^2(1-d_n\lambda_n).
\end{aligned}
$$

To recall the definition of the Fourier coefficients see (ii) in Proposition 2.13. Then, we compute the minimum

$$
\begin{aligned}
&N\hat{k}(j) \\
&= \sum_{t=0}^{N-1}k(t)\exp\left(-\frac{2\pi i}{N}jt\right) \\
&= \exp\left(-\frac{2\pi i}{N}j(N-3)\right)+\gamma\exp\left(-\frac{2\pi i}{N}j(N-2)\right)+\beta\exp\left(-\frac{2\pi i}{N}j(N-1)\right)+1 \\
&= \exp\left(\frac{2\pi i}{N}3j\right)+\gamma\exp\left(\frac{2\pi i}{N}2j\right)+\beta\exp\left(\frac{2\pi i}{N}j\right)+1 \\
&= P\left(\exp\left(\frac{2\pi i}{N}j\right)\right),
\end{aligned}
$$

where we have taken advantage of $\exp(-2\pi ij)=1$ for all $j\in\mathbb{N}$. Thus with

$$
P_{n,\min}^{\pm} := \min_{j=0,\dots,\lfloor 1/\lambda_n\rfloor-3}\frac{1}{2}\left|P^{\pm}\left(\exp\left(\frac{2\pi i}{N}j\right)\right)\right|^2,
$$

we obtain

$$
H_n^{\mathrm{per}}(v)\geq P_{n,\min}(1-d_n\lambda_n).
$$

**Step 3** We conclude the lower bound for $H_n$

$$H_n(u) \geq P_{n,\min}(1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + |\gamma| + |\beta|)^2.$$

We split $H_n$ into bulk and boundary part and make use of the fact, that $c(t) = b(t)$ for $t = 0, \dots, N-4$. Then

$$\begin{aligned}
H_n(u) &= \frac{1}{2}\sum_{t=0}^{N-4} \lambda_n |b(t)|^2 + \frac{1}{2}\sum_{t=N-3}^{N-1} \lambda_n |b(t)|^2 \\
&= \frac{1}{2}\sum_{t=0}^{N-4} \lambda_n |c(t)|^2 + \frac{1}{2}\sum_{t=N-3}^{N-1} \lambda_n |b(t)|^2 \\
&= H_n^{\mathrm{per}}(v) + \frac{1}{2}\sum_{t=N-3}^{N-1} \lambda_n (|b(t)|^2 - |c(t)|^2) \qquad (45) \\
&\geq P_{n,\min}(1 - d_n\lambda_n) - \frac{1}{2}\sum_{t=N-3}^{N-1} \lambda_n |c(t)|^2 \\
&\geq P_{n,\min}(1 - d_n\lambda_n) - \frac{3}{2}\lambda_n \max_{t=N-3,N-2,N-1} |c(t)|^2.
\end{aligned}$$

Using that $|v(t)|^2 = |u^t|^2 = 1$ for all $t = 0, \dots, N-1$ we compute

$$\begin{aligned}
|c(t)|^2 &\leq |v(t + 3 \bmod N) + \gamma v(t + 2 \bmod N) + \beta v(t + 1 \bmod N) + v(t \bmod N)|^2 \\
&\leq (2 + |\gamma| + |\beta|)^2.
\end{aligned}$$

This shows the lower bound for $H_n$. □

**Corollary 5.2** (Bounds for the Infimum of the Energy)**.** *Let $\alpha_1 > 0$, $\alpha_2 > 0$ and $\gamma^\pm, \beta^\pm$ as in (43). Let $E_n^\pm : \bar{\mathcal{U}}_n(I, S^1) \to \mathbb{R}$ for $n \in \mathbb{N}$ be given by*

$$E_n^\pm(u) = \sum_{i=0}^{\lfloor 1/\lambda_n \rfloor - 3} \lambda_n \left(-\alpha_1(u^i, u^{i+1}) + \alpha_2(u^i, u^{i+2}) \pm (u^i, u^{i+3})\right),$$

*as in (40). We define $P^\pm : \mathbb{C} \to \mathbb{C}$ as*

$$P^\pm(z) := z^3 + \gamma^\pm z^2 + \beta^\pm z \pm 1,$$

*and set*

$$P_{n,\min}^\pm := \min_{j=0,\dots,\lfloor 1/\lambda_n \rfloor - 3} \frac{1}{2}\left| P^\pm\left(\exp\left(\frac{2\pi i}{\lfloor 1/\lambda_n \rfloor - 2} j\right)\right)\right|^2.$$

*Then for all $u \in \bar{\mathcal{U}}_n(I, S^1)$ we have*

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n^{\pm} \leq (P_{n,\min}^{\pm} - \omega^{\pm})(1 - d_n\lambda_n),$$
$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n^{\pm} \geq (P_{n,\min}^{\pm} - \omega^{\pm})(1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + |\gamma^{\pm}| + |\beta^{\pm}|)^2.$$

*Proof.* Let $\alpha_1 > 0$ and $\alpha_2 > 0$. Set $N := \lfloor 1/\lambda_n \rfloor - 2$. Let $E_n^{\pm}$ be defined as in (40) and $H_n^{\pm}$ as in (42). The relation in (44) is

$$E_n^{\pm}(u) = H_n^{\pm}(u) - (1 - d_n\lambda_n)\omega^{\pm} \qquad \text{for all } u \in \bar{\mathcal{U}}_n(I, S^1).$$

For legibility reasons we abandon the F/AF distinction ($\pm$) and prove the Corollary for $E_n^+$ only. The proof for $E_n^-$ is analogous.

**Step 1** We start by proving the lower bound.

With Proposition 5.1 we can estimate $H_n$ and therefore $E_n$ for all $u \in \bar{\mathcal{U}}_n(I, S^1)$

$$\begin{aligned} E_n(u) &= H_n(u) - (1 - d_n\lambda_n)\Delta \\ &\geq (P_{n,\min} - \Delta)(1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + |\gamma^{\pm}| + |\beta^{\pm}|)^2 \end{aligned}$$

and thus the estimate holds for the infimum of the energy.

**Step 2** We continue by proving the upper bound.

To show the upper bound we consider functions $u \in \bar{\mathcal{U}}_n(I, S^1)$ that are $N$-periodic, i.e., $u^t = u^{t+N}$ for all $t = 0, \ldots, N+2$. The set of these functions is denoted by $\mathcal{U}_n^{\text{per}}(I, S^1)$ (e.g. $u$ constant is in $\mathcal{U}_n^{\text{per}}(I, S^1)$). This is a stronger assumption then the periodic boundary condition in (41), in particular $\mathcal{U}_n^{\text{per}}(I, S^1) \subseteq \bar{\mathcal{U}}_n(I, S^1)$ and therefore

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n \leq \inf_{\mathcal{U}_n^{\text{per}}(I,S^1)} E_n.$$

In the proof of Proposition 5.1 we take the $j^* := \arg\min_{j=0,\ldots,N-1} \frac{1}{2}\left|P\left(\exp(\frac{2\pi i}{N} j)\right)\right|^2$ and compute the energy of $u_*^t := \exp\left(\frac{2\pi i}{N} j^* t\right)$. Notice $u_* \in \mathcal{U}_n^{\text{per}}(I, \mathbb{C})$ by the identity $\exp(-2\pi i j) = 1$ for all $j \in \mathbb{N}$. Then compute $c(t)$ for $u_*$

$$c(t) = u_*^t P\left(\exp\left(\frac{2\pi i}{N} j^*\right)\right).$$

In the proof the auxiliary sequences $b$ and $c$ coincide on every lattice point for such

periodic functions and hence the boundary term in (45) vanishes, leaving us with

$$\begin{aligned}
H_n(u_*) &= H_n^{\mathrm{per}}(u_*) \\
&= \sum_{t=0}^{N-1} \lambda_n \frac{1}{2} |c(t)|^2 \\
&= \sum_{t=0}^{N-1} \lambda_n \frac{1}{2} \left|u_*^t\right|^2 \left| P\left(\exp\left(\frac{2\pi i}{N} j^*\right)\right)\right|^2 \\
&= \frac{1}{2} \left| P\left(\exp\left(\frac{2\pi i}{N} j^*\right)\right)\right|^2 (1 - d_n\lambda_n) \\
&= P_{n,\min}(1 - d_n\lambda_n).
\end{aligned}$$

We get an upper bound for the infimum of $H_n$

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} H_n \leq P_{n,\min}(1 - d_n\lambda_n).$$

The relation of $E_n$ and $H_n$ in (44) holds true for all $u \in \mathcal{U}_n^{\mathrm{per}}(I, S^1)$, therefore

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n = \inf_{\bar{\mathcal{U}}_n(I,S^1)} H_n - (1 - d_n\lambda_n)\Delta \leq (P_{n,\min} - \Delta)(1 - d_n\lambda_n)$$

and we have shown an upper and lower bound for the infimum of $E_n$. □

**Remark 5.3** (Lower bound of the Energy)**.** *The above results provide two lower bounds. The first one from the relation of the non-negative $H_n^{\pm}$ with $E_n^{\pm}$ (44)*

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n^{\pm} \geq -(1 - d_n\lambda_n)\omega^{\pm}$$

*and the second one given in Corollary 5.2*

$$\inf_{\bar{\mathcal{U}}_n(I,S^1)} E_n^{\pm} \geq (P_{n,\min}^{\pm} - \omega^{\pm})(1 - d_n\lambda_n) - \lambda_n \frac{3}{2}(2 + |\gamma^{\pm}| + |\beta^{\pm}|)^2.$$

*We are going to use the effective lower bound depending on*

$$P_{n,\min}^{\pm}(1 - d_n\lambda_n) \sim \lambda_n \frac{3}{2}(2 + |\gamma^{\pm}| + |\beta^{\pm}|)^2.$$

In Fig. 12 we show the asymptotic behavior of the upper and effective lower bound for the F-AF-AF model with associated energy $E_n^+$. Notice the bounds converge to one and the same limit, as the additional term on the lower bound vanishes as the lattice size increases. Then, Fig. 13 displays numerically minimized energies starting from random initial configurations. While the optimizer occasionally gets trapped in local minima, indicated by the energy spikes, the overall trend of the approximated infimum correctly approaches the theoretical upper bound and respects both bounds.

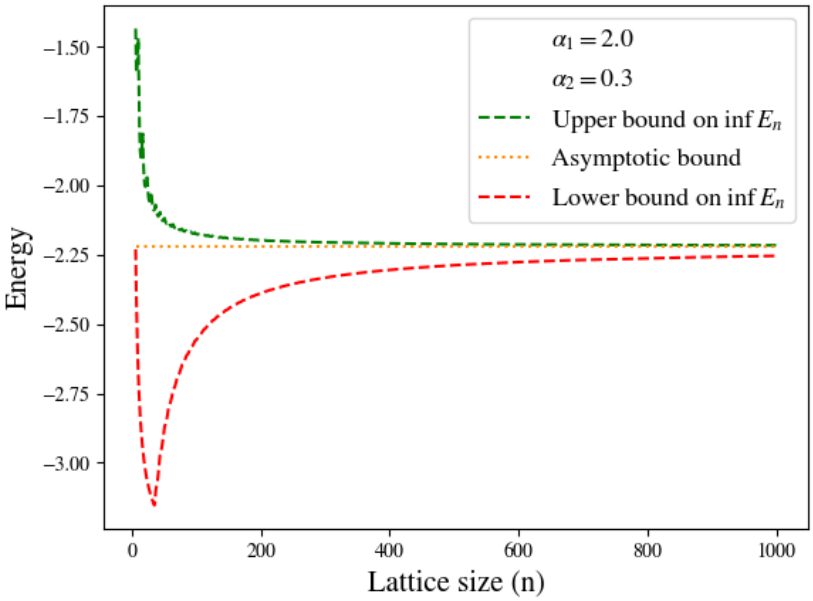


Figure 12: Convergence of the theoretical bounds on the infimum of the energy for the F-AF-AF spin chain model from Corollary 5.2, for lattice sizes $n = 6, \ldots, 1000$. The asymptotic values of the bounds are identical.

This supports the theoretical framework.

The minimum in Corollary 5.2 is taken over discrete lattice points of $P^{\pm}$. We can consider $g^{\pm} : [0, 2\pi) \to [0, +\infty)$, $g^{\pm}(\varphi) := \frac{1}{2}|P^{\pm}(\exp(i\varphi))|^2$ and expand the modulus of $P^{\pm}$

$$\begin{aligned} g^{\pm}(\varphi) &= 1 + \frac{(\gamma^{\pm})^2 + (\beta^{\pm})^2}{2} + \cos(\varphi)(\gamma^{\pm} + \gamma^{\pm}\beta^{\pm} \pm \beta^{\pm}) \\ &\quad + \cos(2\varphi)(\beta^{\pm} \pm \gamma^{\pm}) \pm \cos(3\varphi) \\ &= 1 + \frac{\alpha_2^2}{2} + \alpha_2 \pm \alpha_1 + \cos(\varphi)(-\alpha_1) + \cos(2\varphi)(\alpha_2) \pm \cos(3\varphi). \\ &= \omega^{\pm} + \cos(\varphi)(-\alpha_1) + \cos(2\varphi)(\alpha_2) \pm \cos(3\varphi) \end{aligned}$$

and use this expression to numerically minimize $g^{\pm}$. Further, we show this is equivalent to assuming $u \in \bar{\mathcal{U}}_n(I, S^1)$ such that for $\varphi \in [0, \pi]$

$$u^t := \begin{pmatrix} \cos(\varphi t) \\ \sin(\varphi t) \end{pmatrix}$$

for all $t = 0. \ldots, \lfloor 1/\lambda_n \rfloor$. The energy for such a state simplifies (see computation in the proof of Proposition 4.1), so that we can consider it depending only on the

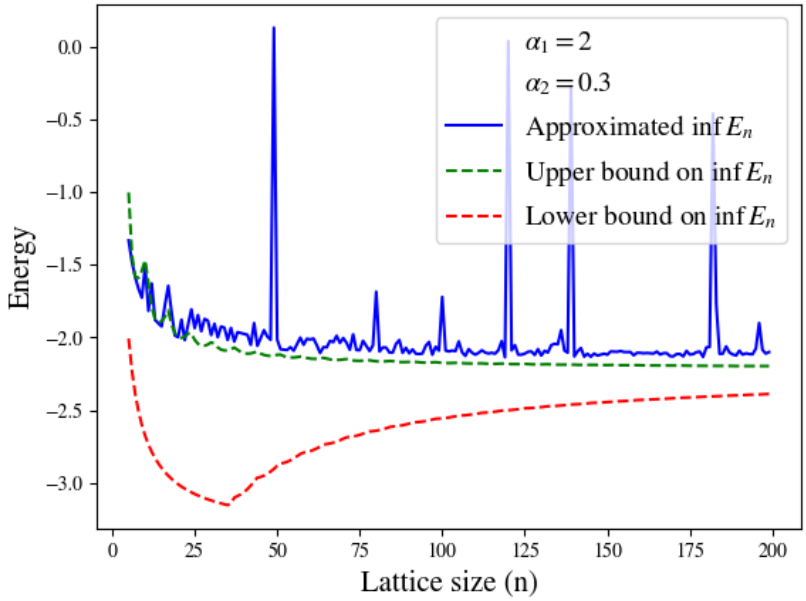


Figure 13: Convergence of the numerically minimized energy for the NNNN model for lattice sizes $n = 5, \ldots, 200$. The numerical result is compared to the theoretical bounds from Corollary 5.2. This suggest existence of local minima and supports the theoretical results, by respecting the bounds.

variable $\varphi$.

$$\begin{aligned}
E_n^{\pm}(u) &= \sum_{t=0}^{N-1} \lambda_n \left(-\alpha_1(u^t, u^{t+1}) + \alpha_2(u^t, u^{t+2}) \pm (u^t, u^{t+3})\right) \\
&= \sum_{t=0}^{N-1} \lambda_n \left(-\alpha_1 \cos(\varphi) + \alpha_2 \cos(2\varphi) \pm \cos(3\varphi)\right) \\
&= \left(-\alpha_1 \cos(\varphi) + \alpha_2 \cos(2\varphi) \pm \cos(3\varphi)\right)(1 - d_n\lambda_n). \\
&= g^{\pm}(\varphi) - \omega^{\pm}
\end{aligned}$$

Motivated by this and the result of Proposition 4.1 and Remark 4.2 we analyze the model for particular states (see Fig. 1). We assume a constant angle $\varphi \in [0, \pi]$ between adjacent spins. This represents helimagnetic, ferromagnetic and antiferromagnetic states of the system. Let $u \in \bar{\mathcal{U}}_n(I, S^1)$ be as above. We are looking for minimizers. The necessary condition for a minimum is

$$0 \overset{!}{=} \frac{\partial E_n^{\pm}(\varphi)}{\partial \varphi} = \sin(\varphi)\left[(\alpha_1 \pm 3) - 4\alpha_2 \cos(\varphi) \mp 12\cos^2(\varphi)\right](1 - d_n\lambda_n),$$

so stationary points are $\varphi_F^{\pm} = 0$, $\varphi_{AF}^{\pm} = \pi$ and

$$\begin{aligned}
\cos(\varphi_H^{+}) &= \frac{-\alpha_2 \pm \sqrt{\alpha_2^2 + 3\alpha_1 + 9}}{6} \\
\cos(\varphi_H^{-}) &= \frac{\alpha_2 \pm \sqrt{\alpha_2^2 - 3\alpha_1 + 9}}{6}.
\end{aligned}$$

$\varphi_F^{\pm}$ corresponds to ferromagnetic states, since spins are aligned, when the angle between them is 0. $\varphi_{AF}^{\pm}$ corresponds to antiferromagnetic states, here spin orientations alternate. $\varphi_H^{\pm}$ correspond to helimagnetic states with constant angle $\varphi_H^{\pm}$ inbetween spins. We concentrate on the F, AF states. Compute the second derivate

$$\frac{\partial^2 E_n^{\pm}(\varphi)}{\partial \varphi^2} = (-\alpha_1 \cos(\varphi) + \alpha_2 \cos(2\varphi) \pm \cos(3\varphi))(1 - d_n\lambda_n)$$

The sufficient condition for a minimum is

$$0 \leq \frac{\partial^2 E_n^{\pm}(\varphi_j^{\pm})}{\partial(\varphi_j^{\pm})^2} = (1 - d_n\lambda_n) \begin{cases} (\alpha_1 - 4\alpha_2 \mp 9) & \text{if } j = F \\ (-\alpha_1 - 4\alpha_2 \pm 9) & \text{if } j = AF. \end{cases} \tag{46}$$

This defines conditions (independent of $n$) for the interaction ratios, where ferromagnetic, respectively antiferromagnet local minima can be found. Since, additionally, we have $\alpha_1 > 0$, $\alpha_2 > 0$ the equation $0 \leq (-\alpha_1 - 4\alpha_2 - 9)$ can never hold, i.e., the antiferromagnetic case is forbidden. Fig. 14 displays the stability regions for the local minima of $E_n^{+}$. Fig. 15 displays the stability regions for the local minima of $E_n^{-}$.

As an example we focus on the F-AF-AF (+) spin chain model. The corresponding

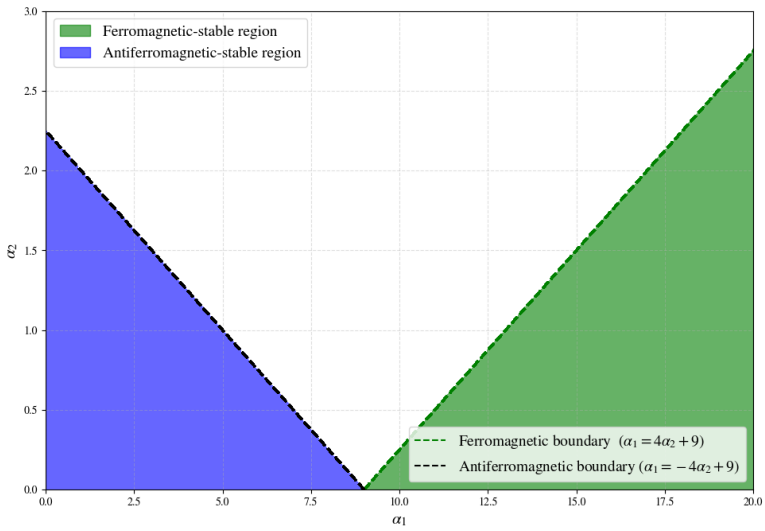


Figure 14: Stability regions for the F-AF-AF (+) spin chain model in the $(\alpha_1, \alpha_2)$-plane (46). Green shading marks parameter pairs where the ferromagnetic configuration is a local minimum, blue shading marks pairs where the antiferromagnetic configuration is a local minimum.

ferromagnetic energies are

$$E^+_{n,F} := E^+_n(\varphi^+_F) = (-\alpha_1 + \alpha_2 + 1)(1 - d_n\lambda_n) \tag{47}$$

In Fig. 16 we compare the ferromagnetic energy to the bounds from Corollary 5.2. This indicates, that ferromagnetic states are candidates for ground states in the F-stability region. Last but not least, Fig. 17 shows the minimization of a random initial spin chain in the ferromagnetic regime. The numerical minimizer yields an energy value close to the optimal ferromagnetic spin chain, which is given as reference. This supports the hypothesis of it being a ground state. This being a promising start for future research, the thesis finalizes by concluding the results.

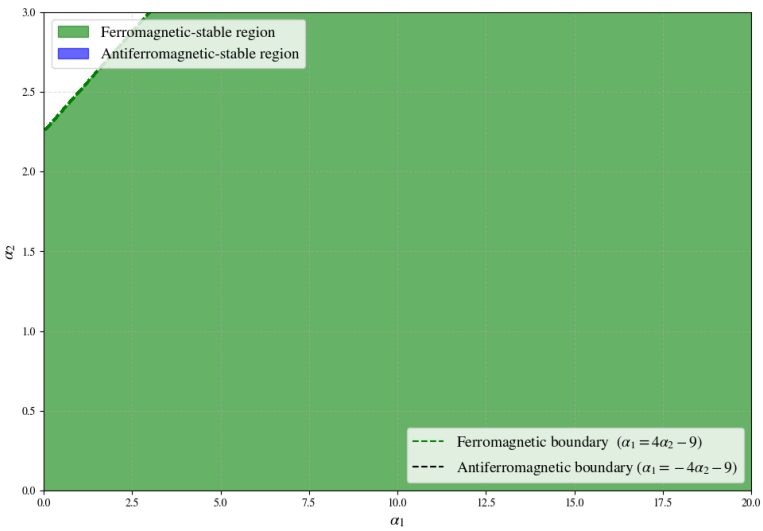


Figure 15: Stability regions for the F-AF-F $(-)$ spin chain model in the $(\alpha_1, \alpha_2)$-plane (46). Green shading marks parameter pairs where the ferromagnetic configuration is a local minimum. The antiferromagnetic values lies entirely in the negative $\alpha_1$-half-plane, so it is outside the physically admissible quadrant.

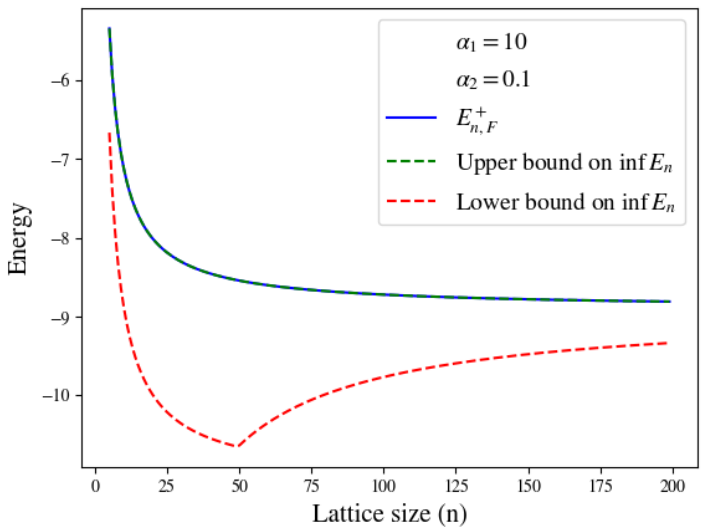


Figure 16: This plot visually demonstrates the convergence of the ferromagnetic energy $E^+_{n,F}$ (47) within the F-stability region for the extended spin chain model. It compares the ferromagnetic energy to the bounds from Corollary 5.2. The convergence of the ferromagnetic energy to the tight upper bound suggests it is indeed the ground state in this parameter regime.

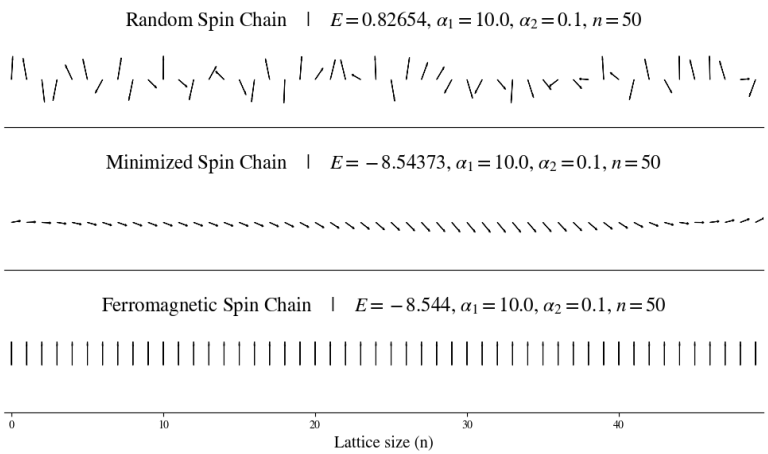


Figure 17: Comparison of spin configurations for lattice size $n = 50$ in the ferromagnetic regime (46) $\alpha_1 = 10$, $\alpha_2 = 0.1$. (Top) Initial random configuration. (Middle) Minimizing configuration resulting from minimization process. (Bottom) Theoretical locally minimizing spin configuration. For reference: the energy value for each spin chain.The energy of the numerically minimized state (Middle) is nearly identical to the theoretical energy of the ferromagnetic state. Notice the minimizer supports the hypothesis, that the ferromagnetic state is the global minimizer by converging from a random state towards the ferromagnetic state.

# 6 Conclusion

At the heart of this thesis stands the thorough analysis of the F-AF model of a one-dimensional frustrated spin chain.

First, in Proposition 4.1 we characterize the minimizers and minima of the finite energy functional describing the model. The findings reveal a helimagnetic ground state for interaction ratios ($0 < \alpha \leq 4$) and a ferromagnetic state for $\alpha \geq 4$. The central theoretical result is the presentation of the $\Gamma-$limit for this energy functional, as detailed in the proof of Theorem 4.4. Of particular interest is the minimum value of the limit energy, proven in Theorem 4.7. This follows as a consequence of an important result from $\Gamma-$convergence theory, which is proven in Theorem 3.12. It turns out that the continuum limit energy corresponds to the results from the discussion of the finite energy, respecting the aforementioned regimes for $\alpha$. Collectively, the numerical analysis of the F-AF model appears consistent with the theoretical results. Notably, the illustration of minimizers demonstrates the helimagnetic and ferromagnetic behavior. The numerical analysis further revealed that random initial spin chains often converge to local minima containing energetically costly chirality transitions. However, the numerical work also highlighted its limitations without a proper analytical foundation, as seen in the tendency of optimizers to converge to local minima and in the approximation of the energy density $f_{\text{hom}}$ from Theorem 4.4.

Second, the thesis explores the helimagnetic/ferromagnetic transition point ($\alpha \simeq 4$) of the model. Heuristics are presented for the $\Gamma-$limit of a rescaled energy functional in Theorem 4.9. Broadly translated, this accounts for the number of energetically costly chirality transitions a spin system undergoes. The limit energy is governed by a scaling parameter $l$, contrasting the convergence of the lattice size and the approach to the transition point. For $l = 0$ the system allows for a finite number of chirality transitions, whereas for $l = +\infty$ the frustration vanishes faster than the lattice scale and such transitions are energetically forbidden.

Finally, the model is extended by considering NNNN interactions. In Corollary 5.2, upper and lower bounds for the infimum of the finite energy functional are proven by analyzing the system in the Fourier space. The bounds differ by a term that vanishes as the lattice size diverges, suggesting that the limit energy coincides with the asymptotic behavior of the bounds. A numerical evaluation of the minimum values of the energy functional supports this hypothesis. Despite the limitations of the analysis, a ferromagnetic local minimum was found which depends on the interaction ratios $\alpha_1$ and $\alpha_2$, presenting a valuable direction for future investigation of ground states.

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# Appendix

## Code Availability

The complete Python codebase used for numerical analysis in this thesis is available at:

`https://github.com/leanderroller/bachelor-thesis-code`

to ensure transparency and reproducibility of the results presented.

## Selbständigkeitserklärung

Ich erkläre hiermit, dass ich die vorliegende Arbeit selbständig verfasst und noch nicht für andere Prüfungen eingereicht habe. Sämtliche Quellen einschließlich Internetquellen, die unverändert oder abgewandelt wiedergegeben werden, insbesondere Quellen für Texte, Grafiken, Tabellen und Bilder, sind als solche kenntlich gemacht. Mir ist bekannt, dass bei Verstößen gegen diese Grundsätze ein Verfahren wegen Täuschungsversuchs bzw. Täuschung eingeleitet wird.

Berlin, den July 12, 2025